

[533] (CONTINUATION)

On Factorials and Derivations.

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p. 467 [533]

where the symbols $()_\theta$ denote sums of products of the differences of an upper letter and a lower letter, θ factors in each product; and the several sums being formed as follows:

$$\text{For } \left(\begin{array}{c} \alpha, \beta, \gamma, \delta, \epsilon \\ a, b, c, d, e \end{array} \right)_1 \text{ write } \begin{array}{c|c} \alpha & a \\ \beta & b \\ \gamma & c \\ \delta & d \\ \epsilon & e; \end{array}$$

the expression is

$$(\alpha - a) + (\beta - b) + (\gamma - c) + (\delta - d) + (\epsilon - e).$$

$$\text{For } \left(\begin{array}{c} \alpha, \beta, \gamma, \delta \\ a, b, c, d, e \end{array} \right)_2 \text{ write } \begin{array}{c|c} \alpha, \alpha & a, b \\ \alpha, \beta & a, c \\ \alpha, \gamma & a, d \\ \beta, \beta & b, c \\ \alpha, \delta & a, e \\ \beta, \gamma & b, d \\ \beta, \delta & b, e \\ \gamma, \gamma & c, d \\ \gamma, \delta & c, e \\ \delta, \delta & d, e; \end{array}$$

viz. the expression is

$$(\alpha - a)(\alpha - b) + (\alpha - a)(\beta - c) + \cdots + (\delta - d)(\delta - e).$$

$$\text{For } \left(\begin{array}{c} \alpha, \beta, \gamma \\ a, b, c, d, e \end{array} \right)_3 \text{ write } \begin{array}{c|c} \alpha, \alpha, \alpha & a, b, c \\ \alpha, \alpha, \beta & a, b, d \\ \alpha, \alpha, \gamma & a, b, e \\ \alpha, \beta, \beta & a, c, d \\ \alpha, \beta, \gamma & a, c, e \\ \beta, \beta, \beta & b, c, d \\ \alpha, \gamma, \gamma & a, d, e \\ \beta, \beta, \gamma & b, c, e \\ \beta, \gamma, \gamma & b, d, e \\ \gamma, \gamma, \gamma & c, d, e; \end{array}$$

viz. the expression is

$$(\alpha - a)(\alpha - b)(\alpha - c) + \cdots + (\gamma - c)(\gamma - d)(\gamma - e).$$

p. 468 [533]

$$\text{For } \left(\begin{array}{c} \alpha, \beta \\ a, b, c, d, e \end{array} \right)_4 \text{ write } \begin{array}{c|c} \alpha, \alpha, \alpha, \alpha & a, b, c, d \\ \alpha, \alpha, \alpha, \beta & a, b, c, e \\ \alpha, \alpha, \beta, \beta & a, b, d, e \\ \alpha, \beta, \beta, \beta & a, c, d, e \\ \beta, \beta, \beta, \beta & b, c, d, e \end{array}.$$

viz. the expression is

$$(\alpha - a)(\alpha - b)(\alpha - c)(\alpha - d) \cdots + (\beta - b)(\beta - c)(\beta - d)(\beta - e).$$

Finally for

$$\left(\begin{array}{c} \alpha \\ a, b, c, d, e \end{array} \right)_5 \text{ write } \alpha, \alpha, \alpha, \alpha, \alpha \mid a, b, c, d, e;.$$

viz. the expression is

$$(\alpha - a)(\alpha - b)(\alpha - c)(\alpha - d)(\alpha - e),$$

which explains the law of the formation of the several coefficients. It is to be observed that in forming the development of any symbol, for instance $\left(\begin{array}{c} \alpha, \beta, \gamma \\ a, b, c, d, e \end{array} \right)_3$, the first column contains the homogeneous products, 3 together, of α, β, γ ; the second column the combinations (that is, combinations without repetitions) 3 together of a, b, c, d, e : the top line is $\alpha, \alpha, \alpha \mid a, b, c$ and to form the subsequent lines we must for any advance α into β , &c. of a greek letter make the like advance a into b , b into c , or c into d , of the corresponding latin letter.

Two particular cases of the theorem may be noticed: if the latin letters all vanish, we have, for example,

$$\begin{aligned} h^5 &= (h - \alpha)(h - \beta)(h - \gamma)(h - \delta)(h - \epsilon) \\ &+ (h - \alpha)(h - \beta)(h - \gamma)(h - \delta) H_1(\alpha, \beta, \gamma, \delta, \epsilon) \\ &+ (h - \alpha)(h - \beta)(h - \gamma) H_2(\alpha, \beta, \gamma, \delta) \\ &+ (h - \alpha)(h - \beta) H_3(\alpha, \beta, \gamma) \\ &+ (h - \alpha) H_4(\alpha, \beta) \\ &+ 1 H_5(\alpha), \end{aligned}$$

where the symbols H denote the sum of the homogeneous products of the annexed letters, taken together according to the suffix number: the last coefficient $H_5(\alpha)$ is of course $= \alpha^5$. And if the greek letters all vanish, then we have in like manner

$$\begin{aligned} (h - a)(h - b)(h - c)(h - d)(h - e) &= h^5 \\ &- h^4 C_1(a, b, c, d, e) \\ &+ h^3 C_2(a, b, c, d, e) \\ &- h^2 C_3(a, b, c, d, e) \\ &+ h C_4(a, b, c, d, e) \\ &- C_5(a, b, c, d, e), \end{aligned}$$

p. 469 [533]

where the symbols C denote the combinations of the annexed letters taken together according to the suffix number; the last coefficient $C_5(a, b, c, d, e)$ is of course $= abcde$. This is the ordinary theorem giving the expression of an equation in terms of its roots.

Combining the two theorems, if in the first theorem we express the products $(h-\alpha)(h-\beta)(h-\gamma)(h-\delta)(h-\epsilon)$, &c. in powers of h by means of the second theorem; or if in the second theorem we express the powers h^5, h^4 , &c. in terms of the products $(h-a)(h-b)(h-c)(h-d)(h-e)$, &c. by means of the first theorem; then in either case we obtain certain identical relations connecting the C, H of $(\alpha, \beta, \dots)$ or of $(a, b, \dots)$.

I have mentioned the factorial notation

$$[m]^r = m(m-1) \cdots (m-r+1),$$

where r is a positive integer; a consequence of this is

$$[m]^{r+s} = [m]^r [m-r]^s,$$

where r and s are positive integers; or as this may also be written

$$[m+r]^{r+s} = [m+r]^r [m]^s.$$

Assuming this to subsist for $s = 0$, or a negative integer; first for $s = 0$, we have $[m+r]^r = [m+r]^r [m]^0$; that is, $[m]^0$ is $= 1$; and then for $s = -r$, we have $1 = [m+r]^r [m]^{-r}$; that is,

$$[m]^{-r} = \frac{1}{[m+r]^r},$$

and in particular, $r = 1, 2$, &c., we have

$$[m]^{-1} = \frac{1}{m+1}, \quad [m]^{-2} = \frac{1}{(m+1)(m+2)}, \quad \&c.,$$

which explains the extension of the factorial notation to negative integer values of the index.

But the equation

$$[m]^{r+s} = [m]^r [m-r]^s$$

does not in any determinate manner lead to an extension of the factorial notation to fractional or other values of the index. In fact, assuming $[m]^r = \frac{\Pi m}{\Pi(m-r)}$, where Π is an arbitrary functional symbol, the equation in question becomes

$$\frac{\Pi m}{\Pi(m-r-s)} = \frac{\Pi m}{\Pi(m-r)} \frac{\Pi(m-r)}{\Pi(m-r-s)},$$

viz. the original equation is identically satisfied, without any condition whatever being imposed upon the function Π , and on this account we have not, in the notation of the factorial with an integer index, any sufficient basis for a theory of general differentiation.

p. 470 [533]

A product

$$m(m - \alpha) \cdots \{m - (r - 1)\alpha\}$$

can of course be expressed in the factorial notation, viz. it is

$$= \alpha^r \left[\frac{m}{\alpha} \right]^r,$$

and on this account it is not in general necessary to employ a notation such as $[m, \alpha]^r$ to denote such a factorial wherein the difference of the successive factors instead of being $= -1$ is $= -\alpha$; in particular cases where factorials of the kind in question are used, it may be convenient to employ such a notation. In particular it is sometimes convenient to use the notation $[m, -1]^r$ or better $\{m\}^r$ to denote the product

$$m(m + 1) \cdots (m + r - 1),$$

where the successive factors instead of being diminished, are increased by unity. It may be noticed, that reversing in this last product the order of the factors, we find

$$\{m\}^r = [m + r - 1]^r;$$

a somewhat similar formula, but employing only the ordinary factorial notation, is obtained from the equation

$$[m]^r = m(m - 1) \cdots (m - r + 1),$$

by first changing the sign of m and then reversing the order of the factors; viz. we have

$$[-m]^r = (-)^r [m + r - 1]^r.$$

Reverting to the process used for the development of the expressions $()_\theta$, where there are two columns, the one of greek, the other of latin letters; it is to be remarked that although the order in which the successive lines are evolved is not material for the purpose of the theorem, yet that a certain definite order of evolution has been made use of; thus in regard to $\left(\begin{smallmatrix} \alpha, \beta, \gamma, \delta \\ a, b, c, d, e \end{smallmatrix} \right)_2$, the column of greek letters, giving the homogeneous products of the second order in $(\alpha, \beta, \gamma, \delta)$, was

$$\begin{array}{cc} \alpha & \alpha \\ \alpha & \beta \\ \alpha & \gamma \\ \beta & \beta \\ \alpha & \delta \\ \beta & \gamma \\ \beta & \delta \\ \gamma & \gamma \\ \gamma & \delta \\ \delta & \delta \end{array}$$

p. 471 [533]

this is evolved from the top term (α, α) by a process given implicitly in Arbogast's Calculus of Derivations, and which may be termed the rule of the last and the last but one. Let the direction "operate on any letter," be understood to mean that the letter in question is to be changed into that which immediately follows it, but in such wise that when the letter occurs more than once, e.g. as in α, α the operation affects only the letter in the right-hand place. Then operate on the α, α in regard to α , we obtain α, β ; operate on this in regard to β , we obtain α, γ ; and in regard to α , we obtain β, β . Again we operate on α, γ in regard to γ , and obtain α, δ ; we do not operate on it in regard to α for the reason that α is not the letter immediately preceding γ . Operate on β, β in regard to β , we obtain β, γ . The next step, if the series extended to ϵ would be to operate on α, δ in regard to δ , giving α, ϵ ; do not operate on it in regard to α , for the reason that α is not the letter immediately preceding δ . But in the example, since the series does not extend to ϵ , there is no operation on α, δ . Passing then to the next term β, γ , we operate in regard to γ , obtaining β, δ , and since β is the letter immediately preceding γ , we also operate in regard to β , obtaining γ, γ . Similarly if ϵ were admissible, β, δ would give β, ϵ , but it in fact gives nothing; γ, γ gives γ, δ ; thus if ϵ were admissible would give γ, ϵ and δ, δ , but it in fact gives only δ, δ , and, ϵ being inadmissible, the process is here concluded. The rule is, operate on the last letter, and when the last but one letter is that which, in alphabetical order, immediately precedes the last letter (but in this case only) operate on the last but one letter.

Taking another example, but with numbers instead of letters, and supposing the highest admissible number to be 5, then from 111 we derive as follows:

111	112	113	114	115	125	135	145	155	255	355	455	555
		122	123	124	134	144	235	245	345	445		
			222	133	224	225	244	335	444			
				223	233	234	334	344				
						333						

the original single column being here for greater convenience broken up into distinct columns; but the order of the terms, when the columns are taken one after the other in order, each being read from the top to the bottom, being the same as before; it will be noticed that the successive divisions are the partitions into 3 parts (no part exceeding 5) of the numbers 3, 4, ..., 15 respectively; the partitions being in each case obtained without repetition, and those of the same number being given, say in their numerical order (corresponding with the alphabetical order when letters are employed). It is necessary to show that the partitions will be obtained without repetitions; and that all the partitions will be obtained; for this purpose consider, for example, the partitions of 9; any one of these is either a partition 135 where the last number 5 is not a repeated number; and in this case there is a partition of 8, viz. 134, from which operating on the last we obtain 135; but there is no other partition of 8 which would give 135, the only such partition would be 125, but here, as 2 is not the number which immediately precedes 5, there is no operation on the last but one, and we do not from it obtain 135. Or else a partition of 9 is of the form 144

p. 472 [533]

where the last letter is repeated; there exists in this case no partition of 8, such that operating on the last we obtain from it 144, but there does exist a partition of 8, viz. 134, such that operating on the last but one we obtain from it 144. That is, for any partition whatever of 9 there exists one (and only one) partition of 8, such that operating on the last or the last but one, we obtain from it the partition of 9; that is, taking the entire system of the partitions of 8, and operating on the last and the last but one, we obtain, and that without repetitions, the entire series of the partitions of 9; and so in general.

Translating the example into letters, but using for greater convenience a^2 , &c. instead of a, a , &c. the process will be precisely the same; taking the letters to be a, b, c, d, e , we have

$$\begin{array}{cccccccccccccc}
 a^3 & a^2b & a^2c & a^2d & a^2e & abe & ace & ade & ae^2 & be^2 & ce^2 & de^2 & e^3 \\
 & ab^2 & abc & abd & acd & ad^2 & bce & bde & cde & d^2e & & & \\
 & & b^3 & ac^2 & b^2d & b^2e & bd^2 & c^2e & d^3 & & & & \\
 & & & b^2c & bc^2 & bcd & c^2d & cd^2 & & & & & \\
 & & & & & c^3 & & & & & & &
 \end{array}$$

Attributing weights to the several letters, viz. to a, b, c, d, e the weights 1, 2, 3, 4, 5 respectively, the several columns show the terms of the weights 3, 4, ..., 15 respectively.

I have said that the foregoing rule is given implicitly in Arbogast's Calculus of Derivations; this calculus includes in fact a process for the expansion of a function

$$\phi(a + bx + cx^2 + dx^3 + \&c.)$$

in powers of x ; the expansion in question may be obtained by means of Taylor's theorem, viz.

$$\begin{aligned}
 \phi(a + bx + cx^2 + dx^3 + \&c.) &= \phi a \\
 &+ \frac{\phi' a}{1}(bx + cx^2 + dx^3 + \&c.) \\
 &+ \frac{\phi'' a}{1 \cdot 2}(bx + cx^2 + dx^3 + \&c.)^2 \\
 &+ \frac{\phi''' a}{1 \cdot 2 \cdot 3}(bx + cx^2 + dx^3 + \&c.)^3,
 \end{aligned}$$

viz. expanding the several powers of the polynomial increment, and arranging in powers of x , this is

$$\begin{aligned}
 &= \phi a \\
 &+ x(\phi' a \cdot b) \\
 &+ x^2 \left(\phi' a \cdot c + \phi'' a \cdot \frac{b^2}{2} \right) \\
 &+ x^3 \left(\phi' a \cdot d + \phi'' a \cdot bc + \phi''' a \cdot \frac{b^3}{6} \right) \\
 &+ x^4 \left\{ \phi' a \cdot e + \phi'' a \cdot \left(bd + \frac{1}{2}c^2 \right) + \phi''' a \cdot \frac{b^2c}{2} + \phi'''' a \cdot \frac{b^4}{24} \right\} \\
 &+ \&c..
 \end{aligned}$$

p. 473 [533]

but the object of the rule is to obtain this last-mentioned result directly, and understanding “operate in regard to a letter” to mean differentiate with regard to this letter and integrate with respect to the next succeeding letter, then the coefficients of the successive powers of x are all obtained from the first coefficient ϕa , by operating thereon according to the rule of the last and the last but one; thus ϕa , operating on a gives $\phi' a \cdot b$; this operating in regard to b gives $\phi' a \cdot c$, and in regard to a gives

$$\phi'' a \cdot \frac{b^2}{2};$$

the term $\phi' a \cdot c$ is to be operated upon in regard to c only, and it gives

$$\phi' a \cdot d;$$

the other term $\phi'' a \cdot \frac{b^2}{2}$ operated on in regard to b gives $\phi'' a \cdot bc$, and in regard to a it gives $\phi''' a \cdot \frac{b^3}{6}$;

and so on. But attending only to the literal parts, the terms, for instance $\frac{b^2}{c^2}, bc, bd, \&c.$, which present themselves in the formula, are the homogeneous terms derived from b^2 , by the rule, as originally stated, with a view to the derivation of such terms.

p. 476 [534]

Hence assuming

$$\frac{\theta + \alpha}{\phi + \alpha} = \lambda,$$

we may write

$$\frac{\theta + \alpha}{\phi + \alpha} = \lambda, \quad \frac{\theta + \beta}{\phi + \beta} = -\lambda, \quad \frac{\theta + \gamma}{\phi + \gamma} = i\lambda, \quad \frac{\theta + \delta}{\phi + \delta} = -i\lambda,$$

$$\{i = \sqrt{-1} \text{ as usual}\},$$

viz. this is one of three systems of equations; the other two may be obtained therefrom by writing γ, δ, β and δ, β, γ successively in place of β, γ, δ . Hence assuming

$$v = \frac{\theta + u}{\phi + u},$$

the four values of u are $\alpha, \beta, \gamma, \delta$, and the corresponding four values of v are $\lambda, -\lambda, i\lambda, -i\lambda$; and v, u are linearly related to each other; the anharmonic ratio of $(\alpha, \beta, \gamma, \delta)$ is therefore equal to that of $(1, -1, i, -i)$, viz. we have

$$\frac{(\alpha - \gamma)(\beta - \delta)}{(\alpha - \delta)(\beta - \gamma)} = \frac{(1 - i)(-1 + i)}{(1 + i)(-1 - i)} = \frac{(1 - i)^2}{(1 + i)^2} = -1,$$

that is,

$$(\alpha - \gamma)(\beta - \delta) + (\alpha - \delta)(\beta - \gamma) = 0,$$

or, what is the same thing,

$$2(\alpha\beta + \gamma\delta) - (\alpha + \beta)(\gamma + \delta) = 0,$$

viz. we have this relation, or else one of the like relations

$$2(\alpha\gamma + \delta\beta) - (\alpha + \gamma)(\delta + \beta) = 0,$$

$$2(\alpha\delta + \beta\gamma) - (\alpha + \delta)(\beta + \gamma) = 0,$$

that is, the product of the three functions $2(\alpha\beta + \gamma\delta) - (\alpha + \beta)(\gamma + \delta)$ is $= 0$.

But the product in question is (save as to a numerical factor) the cubinvariant J of the quartic function; or the equation in question is the required equation $J = 0$.

More simply, the linear transformation

$$v = \frac{\theta + u}{\phi + u},$$

gives for v the equation $v^4 - \lambda^4 = 0$; which is $(1, 0, 0, 0, -\lambda^4)(v, 1)^4 = 0$; the cubinvariant hereof is $= 0$, and therefore also the cubinvariant of the original function $(a, b, c, d, e)(u, 1)^4$.

Reverting to the equations

$$\frac{\theta + \alpha}{\phi + \alpha} = \lambda, \quad \frac{\theta + \beta}{\phi + \beta} = -\lambda, \quad \frac{\theta + \gamma}{\phi + \gamma} = i\lambda, \quad \frac{\theta + \delta}{\phi + \delta} = -i\lambda,$$

(which, as we have seen, give $2(\alpha\beta + \gamma\delta) = (\alpha + \beta)(\gamma + \delta)$), the same equations give

$$\frac{\theta + \alpha}{\phi + \alpha} + \frac{\theta + \beta}{\phi + \beta} = 0; \quad \frac{\theta + \gamma}{\phi + \gamma} + \frac{\theta + \delta}{\phi + \delta} = 0,$$

p. 477 [534]

that is,

$$\begin{aligned} 2\theta\phi + 2\alpha\beta - (\theta + \phi)(\alpha + \beta) &= 0, \\ 2\theta\phi + 2\gamma\delta - (\theta + \phi)(\gamma + \delta) &= 0, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} 2\theta\phi : 2 : \theta + \phi &= -\alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) \\ &: \gamma + \delta - \alpha - \beta \\ &: \gamma\delta - \alpha\beta, \end{aligned}$$

viz. we have thus the values of $\theta\phi$, $\theta + \phi$ (and thence of θ, ϕ) corresponding to the relation $2(\alpha\beta + \gamma\delta) = (\alpha + \beta)(\gamma + \delta)$ of the roots. And by cyclically permuting β, γ, δ as before, we have the values of $\theta\phi$, $\theta + \phi$ corresponding to the other two forms respectively of the relation between the roots.

3. *If in a plane A, B, C, D are fixed points and P a variable point, find the linear relation*

$$\alpha.PAB + \beta.PBC + \gamma.PCD + \delta.PDA = 0$$

which connects the areas of the triangles PAB , &c.

Taking $(x, y, 1), (x_1, y_1, 1)$, &c. for the coordinates of P, A, B, C, D respectively, we have

$$\begin{aligned} PAB &= \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 012, \quad \text{suppose,} \\ PBC &= 023, \quad \&c. \end{aligned}$$

(where the values of the several determinants fix the signs of the several triangles). The identical equation then is

$$\alpha.012 + \beta.023 + \gamma.034 + \delta.041 = 0;$$

(that such an equation exists appears at once by the consideration that $\alpha, \beta, \gamma, \delta$ can be determined so that the coefficients of x, y , and the constant term shall severally vanish); and in order actually to find the values we may make P coincide with the points A, B, C, D successively. We thus have

$$\begin{aligned} \beta.123 + \gamma.134 &= 0, \\ \gamma.234 + \delta.241 &= 0, \\ \delta.341 + \alpha.312 &= 0, \\ \alpha.412 + \beta.423 &= 0, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} \beta.123 + \gamma.341 &= 0, \\ \gamma.234 + \delta.412 &= 0, \\ \delta.341 + \alpha.123 &= 0, \\ \alpha.412 + \beta.234 &= 0, \end{aligned}$$

p. 478 [534]

and these are at once seen to give

$$\alpha : \beta : \gamma : \delta = 234.341 : -341.412 : 412.123 : -123.341,$$

so that the required identical relation is

$$012.234.341 - 023.341.412 + 034.412.123 - 041.123.341 = 0,$$

in which 012, 023, 034, 041 stand for the triangles PAB, PBC, PCD, PDA , and 234, 341, 412, 123 for the triangles BCD, CDA, DAB, ABC respectively.

4. Find at any point of a plane curve the angle between the normal and the line drawn from the point to the centre of the chord parallel and indefinitely near to the tangent at the point.

Examine whether a like question applies to a point on a surface and the indicatrix section at such point.

Taking the origin at the point on the curve, the axis of x coinciding with the tangent and that of y with the normal; the equation of the curve taken to terms of the third order in x will be

$$y = bx^2 + cx^3,$$

and if, considering x as a small quantity of the first order, and therefore y as a small quantity of the second order, we regard y as given, and find the two values x_1, x_2 , each of the order $\sqrt{y}$, which satisfy the equation, then, as will appear, $x_1 + x_2$ is a small quantity of the order x^2 , and consequently $\frac{x_1 + x_2}{y}$ will have a finite value. And if ϕ be the required angle, then obviously

$$\tan \phi = \frac{\frac{1}{2}(x_1 + x_2)}{y}.$$

We have as a first approximation $bx^2 = y$, or say $x = \pm y^{1/2}/b^{1/2}$, whence to a second approximation

$$bx^2 = y - \frac{cy^{3/2}}{b^{3/2}}, \quad x^2 = \frac{y}{b} \left(1 - \frac{cy^{1/2}}{b^{3/2}} \right),$$

whence

$$x = \frac{y^{1/2}}{b^{1/2}} \left(1 - \frac{cy^{1/2}}{2b^{3/2}} \right) = \frac{y^{1/2}}{b^{1/2}} - \frac{cy}{2b^2};$$

say we have

$$x_1 = \frac{y^{1/2}}{b^{1/2}} - \frac{cy}{2b^2}, \quad x_2 = -\frac{y^{1/2}}{b^{1/2}} - \frac{cy}{2b^2},$$

and thence

$$\frac{1}{2}(x_1 + x_2) = -\frac{cy}{2b^2};$$

whence

$$\tan \phi = -\frac{c}{2b^2},$$

which gives the value of the angle ϕ ; it would be easy to express b, c in terms of the differential coefficients $d_x y, d_x^2 y, d_x^3 y$.

p. 479 [534]

It would at first sight appear that a like question might be asked as to a surface; viz. that it might be proposed to determine the angle between the normal and a line drawn from the point to the centre of the indicatrix conic. But this is not so; in fact, taking the origin at a point on the surface, the axes of x, y being in the tangent plane, and the axis of z coinciding with the normal: then to the third order we have

$$z = (A, B, C)(x, y)^2 + (a, b, c, d)(x, y)^3;$$

but here, regarding z as a given constant, if we take account of the terms of the third order, the section is not a conic but a cubic; and it has not in general any centre; and if (as in the ordinary theory) we neglect the terms of the third order, thus obtaining an indicatrix conic, the centre of this conic lies on the normal, and there is no angle corresponding to the angle ϕ of the plane problem.

The only case where there is such an angle is when the cubic terms $(a, b, c, d)(x, y)^3$ contain as a factor the quadric terms $(A, B, C)(x, y)^2$ (one relation between the coefficients A, B, C, a, b, c, d). For then we have

$$\begin{aligned} z &= (A, B, C)(x, y)^2(1 + 2lx + 2my), \quad \text{viz.} \\ z &= (A, B, C)(x, y)^2 + 2(lxz + myz), \end{aligned}$$

approximately to the third order; and then regarding z as a given constant, this last equation represents a conic having for the coordinates of its centre, say $x = \alpha z, y = \beta z$, and there is an angle $\phi = \tan^{-1} \sqrt{\alpha^2 + \beta^2}$; this is, in fact, what happens in the case of a quadric surface, for the section by a plane parallel and indefinitely near to the tangent plane is then a conic, the centre of which is not on the normal; and the angle ϕ (in the case of a central surface) is in fact the inclination of the normal to the radius from the centre.

I take the opportunity of adding a remark that the indicatrix is never a parabola, but in the separating case between the ellipse and the hyperbola it is a pair of parallel lines. The indicatrix, a parabola, is commonly obtained as follows: viz. taking the axes as before, but starting from an equation $U = 0$, the equation presents itself in the form

$$z = (A, B, C, F, G, H)(x, y, z)^2,$$

which, considering z as a given constant, represents a conic which, it is said, may be a parabola. But observe that z is of the order $(x, y)^2$, the terms $2Fyz + Gzx$, are consequently of the order $(x, y)^3$, but they are not all the terms of this order which would be obtained by the expansion of z as a function of (x, y) ; there is consequently no meaning in retaining them, and they ought to be rejected; similarly the term in z^2 which is of the order $(x, y)^4$ ought to be rejected; the equation is thus reduced to

$$z = Ax^2 + 2Hxy + By^2,$$

which, when $AB - H^2 = 0$, represents not a parabola but a pair of parallel lines. On referring to Dupin's *Développements de Géométrie*, &c. (see p. 49) I find that he is quite accurate; his expression is, "elle peut cependant être une parabole; alors elle se présente sous la forme de deux droites parallèles équidistantes de leur centre": and he afterwards examines in particular "ce cas remarquable."

p. 480 [534]

5. *Shew that a cubic surface has at most four conical points; and a quartic surface at most sixteen conical points.*

If a cubic surface has two conical points, then the line joining these has with the surface two intersections at each of the conical points, and therefore lies wholly in the surface. Hence, for a cubic surface with three conical points A, B, C , the lines AB, BC, CA lie wholly in the surface, and these three lines form the complete section of the surface by the plane ABC : it is clear that there cannot be in this plane a fourth conical point: but there may be, not in this plane, a fourth conical point D . Suppose that this is so, there cannot be a fifth conical point E ; for if there were, the line DE would lie wholly in the surface, and would therefore meet the plane ABC at some point in the section of the surface by this plane; that is, at some point in one of the lines AB, AC, BC ; say at a point in AB : but then the lines AB, DE would intersect, or the four conical points A, B, D, E would lie in a plane. Hence there cannot be any fifth conical point E .

For a quartic surface; suppose this has k conical points, and let any one of these be made the vertex of a cone circumscribing the surface; each generating line is a tangent of the surface; and considering any section by a plane through the vertex, and observing that from a double point of a quartic curve we may draw six tangents to the curve, it appears that the order of the cone is $= 6$. It is easy to see that the lines from the vertex to the remaining $(k - 1)$ conical points are each of them a double line of the cone, and that the cone has not any other double lines; the cone is therefore a cone of the order 6, with $(k - 1)$ double lines. A proper cone of the order 6 has at most 10 double lines, but the cone need not be a proper one; it may, in fact, break up into 6 planes, and in this case the double lines are the 15 lines of intersections of the several pairs of planes. Hence $k - 1$ is $= 15$ at most: or k is $= 16$ at most.

6. *Find the differential equation of the parallel surfaces of an ellipsoid.*

Let (x, y, z) be the coordinates of a point on the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1;$$

(X, Y, Z) the coordinates of a point on the normal at a distance $= k$ from the first-mentioned point. We have

$$\frac{X - x}{x/a^2} = \frac{Y - y}{y/b^2} = \frac{Z - z}{z/c^2} = \rho \quad \text{suppose;}$$

that is,

$$X = x \left(1 + \frac{\rho}{a^2}\right), \quad Y = y \left(1 + \frac{\rho}{b^2}\right), \quad Z = z \left(1 + \frac{\rho}{c^2}\right),$$

and thence

$$k^2 = \rho^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right).$$

Moreover

$$x = \frac{a^2 X}{a^2 + \rho}, \quad y = \frac{b^2 Y}{b^2 + \rho}, \quad z = \frac{c^2 Z}{c^2 + \rho},$$

p. 481 [534]

substituting these values in the equation of the ellipsoid, we have

$$1 = \frac{a^2 X^2}{(a^2 + \rho)^2} + \frac{b^2 Y^2}{(b^2 + \rho)^2} + \frac{c^2 Z^2}{(c^2 + \rho)^2},$$

which determines ρ as a function of X, Y, Z . The tangent plane of the ellipsoid at the point (x, y, z) and of the parallel surface at the point (X, Y, Z) , are parallel to each other (or what is the same thing, the parallel surface cuts at right angles the normal of the ellipsoid), we have therefore

$$\frac{x}{a^2} dX + \frac{y}{b^2} dY + \frac{z}{c^2} dZ = 0,$$

or substituting for x, y, z their values, this is

$$\frac{X dX}{a^2 + \rho} + \frac{Y dY}{b^2 + \rho} + \frac{Z dZ}{c^2 + \rho} = 0,$$

where ρ denotes as above a function of (X, Y, Z) given by the equation

$$1 = \frac{a^2 X^2}{(a^2 + \rho)^2} + \frac{b^2 Y^2}{(b^2 + \rho)^2} + \frac{c^2 Z^2}{(c^2 + \rho)^2}.$$

We have thus the differential equation of the parallel surfaces. It may be remarked, that the integral equation (involving k as the constant of integration), is found by the elimination of x, y, z, ρ from the foregoing equations

$$x = \frac{a^2 X}{a^2 + \rho}, \quad y = \frac{b^2 Y}{b^2 + \rho}, \quad z = \frac{c^2 Z}{c^2 + \rho},$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad k^2 = \rho^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right),$$

or, what is the same thing, by the elimination of ρ from the equations

$$\frac{k^2}{\rho^2} = \frac{X^2}{(a^2 + \rho)^2} + \frac{Y^2}{(b^2 + \rho)^2} + \frac{Z^2}{(c^2 + \rho)^2},$$

$$1 = \frac{a^2 X^2}{(a^2 + \rho)^2} + \frac{b^2 Y^2}{(b^2 + \rho)^2} + \frac{c^2 Z^2}{(c^2 + \rho)^2};$$

these may be replaced by

$$\frac{X^2}{a^2 + \rho} + \frac{Y^2}{b^2 + \rho} + \frac{Z^2}{c^2 + \rho} - \frac{k^2}{\rho} - 1 = 0,$$

$$\frac{X^2}{(a^2 + \rho)^2} + \frac{Y^2}{(b^2 + \rho)^2} + \frac{Z^2}{(c^2 + \rho)^2} - \frac{k^2}{\rho^2} = 0,$$

or, since here the second equation is the derived equation of the first in regard to the parameter ρ , the parallel surface is the envelope of the quadric surface

$$\frac{X^2}{a^2 + \rho} + \frac{Y^2}{b^2 + \rho} + \frac{Z^2}{c^2 + \rho} - \frac{k^2}{\rho} = 0,$$

where ρ is the variable parameter. Or analytically, we find the equation by equating to zero the discriminant in regard to ρ , of the quartic function

$$\rho(a^2 + \rho)(b^2 + \rho)(c^2 + \rho) \left(1 + \frac{k^2}{\rho} - \frac{X^2}{a^2 + \rho} - \frac{Y^2}{b^2 + \rho} - \frac{Z^2}{c^2 + \rho} \right).$$

p. 482 [534]

7. Explain wherein consists the peculiarity of the following problem, and solve it by geometrical considerations:—

Determine the least circle inclosing three given points.

The peculiarity of the problem is that the variable parameters upon which the circle depends, (say α, β the coordinates of the centre and k the radius), are not subject to any equations, but only to the inequalities

$$\begin{aligned} k^2 &> (\alpha - \alpha_1)^2 + (\beta - \beta_1)^2, \\ k^2 &> (\alpha - \alpha_2)^2 + (\beta - \beta_2)^2, \\ k^2 &> (\alpha - \alpha_3)^2 + (\beta - \beta_3)^2, \end{aligned}$$

($\alpha_1, \beta_1; \alpha_2, \beta_2; \alpha_3, \beta_3$, the coordinates of the given points, and the sign $>$ including $=$). The problem therefore cannot be solved by the ordinary analytical method, but it is easily solved geometrically as follows: Let A, B, C be the three points; consider all the circles inclosing the three points, viz. O a circle not passing through any of them; A a circle through the point A , B a circle through the point B , AB a circle through the points A and B , &c. Then for any circle O , if the centre be fixed and the radius gradually diminish, the circle will at last pass through one of the points ABC ; that is, every circle O is greater than some circle A, B , or C ; and the circle O is therefore not a minimum. Taking next a circle A , we may imagine the centre to move from its original position in a straight line towards the point A , the circle thus gradually diminishing until it passes through one of the points B or C ; that is, every circle A is greater than some circle AB or AC , and therefore no circle A is a minimum; and in like manner no circle B or C is a minimum. There remain the circles AB, AC, BC ; if the triangle ABC is acute-angled, then in each series, the least circle is the circle ABC circumscribed about the triangle; and this is then the minimum circle inclosing the three points. But if the triangle is obtuse-angled, say at C , then the least circle CA or CB is the circle ABC circumscribed about the triangle; but this is not the least circle AB , viz. the circle AB , being diminished to ABC , may be further diminished until it becomes the circle on the diameter AB ; but below this it cannot be diminished; and consequently the minimum circle inclosing the three points is in this case the circle on the diameter AB .

8. A particle describes an ellipse under the simultaneous action of given central forces, each varying as (distance)⁻², at the two foci respectively: find the differential relation between the time and the eccentric anomaly.

Taking the equation of the ellipse to be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and the absolute forces at the two foci ($ae, 0$), ($-ae, 0$) to be μ, μ' respectively, the differential equations of motion will be

$$\begin{aligned} \frac{d^2x}{dt^2} &= -\mu \frac{x - ae}{(a - ex)^3} - \mu' \frac{x + ae}{(a + ex)^3}, \\ \frac{d^2y}{dt^2} &= -\mu \frac{y}{(a - ex)^3} - \mu' \frac{y}{(a + ex)^3}. \end{aligned}$$

p. 483 [534]

But if u be the eccentric anomaly, then

$$x = a \cos u, \quad y = b \sin u, = a \sqrt{1 - e^2} \sin u,$$

and the equations become

$$\begin{aligned} -\sin u \frac{d^2 u}{dt^2} - \cos u \left(\frac{du}{dt} \right)^2 &= -\frac{\mu}{a^3} \frac{\cos u - e}{(1 - e \cos u)^3} - \frac{\mu'}{a^3} \frac{\cos u + e}{(1 + e \cos u)^3}, \\ \cos u \frac{d^2 u}{dt^2} - \sin u \left(\frac{du}{dt} \right)^2 &= -\frac{\mu}{a^3} \frac{\sin u}{(1 - e \cos u)^3} - \frac{\mu'}{a^3} \frac{\sin u}{(1 + e \cos u)^3}, \end{aligned}$$

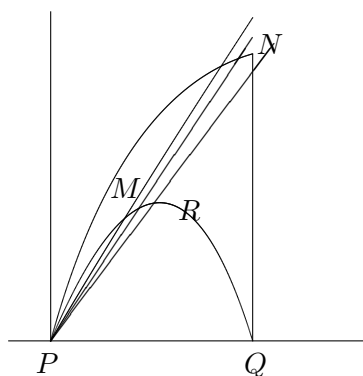
and multiplying by $-\cos u, -\sin u$ respectively, and adding, we have

$$\left(\frac{du}{dt} \right)^2 = \frac{\mu}{a^3} \frac{1}{(1 - e \cos u)^2} + \frac{\mu'}{a^3} \frac{1}{(1 + e \cos u)^2},$$

which is the required differential relation.

9. Show that the attraction of an indefinitely thin double-convex lens on a point at the centre of one of its faces is equal to that of the infinite plate included between the tangent plane at the point and the parallel tangent plane of the other face of the lens.

The figure represents the upper half only of the lens, but in speaking of any portion thereof, such as PRQ , we include the symmetrically situate portion of the under-half of the lens.



Let $\alpha, = PQ$, be the thickness of the lens, $\angle NPQ = \lambda$, which angle is ultimately $= \frac{\pi}{2}$. Then it is at once seen that the attraction of the cone NPQ is $= 2\pi\alpha(1 - \cos \lambda)$: and from this it follows that the attraction of the infinite plate is $= 2\pi\alpha$. The attraction of the whole infinite plate except the cone NPQ is $= 2\pi\alpha \cos \lambda$, which is indefinitely small in regard to $2\pi\alpha$; and, *a fortiori*, the attraction of the portion MPR of the lens is indefinitely small in regard to $2\pi\alpha$. We have then only to show that the attraction of the solid NRQ is indefinitely small in regard to $2\pi\alpha$; for, this being so, the attraction of the lens may be taken to be equal to that of the cone NPQ , and will therefore ultimately be $= 2\pi\alpha$, the attraction of the infinite plate.

p. 484 [534]

Let the position of an element of the solid in question be determined by r its distance from P , θ the inclination of r to the axis PQ , and ϕ the azimuth in regard to any fixed plane through the axis; then $dm = r^2 \sin \theta dr d\theta d\phi$, and the attraction in the direction PQ is

$$\int \sin \theta \cos \theta dr d\theta d\phi = 2\pi \int \left(\frac{\alpha}{\cos \theta} - r \right) \sin \theta \cos \theta d\theta,$$

the integral in regard to ϕ having been taken from $\phi = 0$ to $\phi = 2\pi$, and that in regard to r from $r = r$ (value at the face MQ of the lens) to $r = \frac{\alpha}{\cos \theta}$ (value at the tangent plane QN). Taking the radius of the surface QM of the lens to be $= 1$, we have

$$(1 - \alpha + r \cos \theta)^2 + r^2 \sin^2 \theta = 1,$$

that is,

$$r^2 + 2r \cos \theta (1 - \alpha) = 2\alpha - \alpha^2,$$

or

$$\begin{aligned} \{r + (1 - \alpha) \cos \theta\}^2 &= (1 - \alpha)^2 \cos^2 \theta + 2\alpha - \alpha^2, \\ r &= -(1 - \alpha) \cos \theta + \sqrt{(1 - \alpha)^2 \cos^2 \theta + 2\alpha - \alpha^2}, \end{aligned}$$

which is the value of r to be substituted in the formula

$$\frac{1}{2\pi} A = \int (\alpha \sin \theta - r \sin \theta \cos \theta) d\theta,$$

and the integral is to be taken from $\theta = 0$ to $\theta = \lambda$; viz. this is

$$\begin{aligned} &\int \left[\alpha \sin \theta + (1 - \alpha) \sin \theta \cos^2 \theta - \sin \theta \cos \theta \sqrt{(1 - \alpha)^2 \cos^2 \theta + 2\alpha - \alpha^2} \right] d\theta, \\ &= -\alpha \cos \theta - \frac{1}{3} (1 - \alpha) \cos^3 \theta + \frac{1}{3(1 - \alpha)^2} \{ (1 - \alpha)^2 \cos^2 \theta + 2\alpha - \alpha^2 \}^{3/2}; \end{aligned}$$

so that taking this between the limits in question, we have

$$\frac{1}{2\pi} A = \alpha(1 - \cos \lambda) + \frac{1}{3} (1 - \alpha)(1 - \cos^3 \lambda) + \frac{1}{3(1 - \alpha)^2} \left[\{ (1 - \alpha)^2 \cos^2 \lambda + 2\alpha - \alpha^2 \}^{3/2} - 1 \right],$$

or writing for greater convenience $\lambda = \frac{\pi}{2} - \mu$, ($\mu = \angle PNQ$), this is

$$\begin{aligned} \frac{1}{2\pi} A &= \alpha(1 - \sin \mu) + \frac{1}{3} (1 - \alpha)(1 - \sin^3 \mu) \\ &\quad + \frac{1}{3(1 - \alpha)^2} \left[\{ (1 - \alpha)^2 \sin^2 \mu + 2\alpha - \alpha^2 \}^{3/2} - 1 \right] \\ &= \alpha + \frac{1}{3} \left\{ 1 - \alpha - \frac{1}{(1 - \alpha)^2} \right\} - \alpha \sin \mu \\ &\quad + \frac{1}{3(1 - \alpha)^2} \left[\{ (1 - \alpha)^2 \sin^2 \mu + 2\alpha - \alpha^2 \}^{3/2} - (1 - \alpha)^3 \sin^3 \mu \right] \\ &= -\frac{1}{3(1 - \alpha)^2} (-3\alpha^2 + 2\alpha^3) - \alpha \sin \mu \\ &\quad + \frac{1}{3(1 - \alpha)^2} \left[\{ (1 - \alpha)^2 \sin^2 \mu + 2\alpha - \alpha^2 \}^{3/2} - (1 - \alpha)^3 \sin^3 \mu \right]; \end{aligned}$$

p. 485 [534]

$\sin \mu$ is here an indefinitely small quantity of the order $\alpha^{1/2}$, all the terms are therefore at least of the order $\alpha^{3/2}$, and are to be neglected in comparison with α ; or neglecting such terms we have $A = 0$ (that is, the attraction of the solid NRQ is indefinitely small in regard to α); and the theorem is thus proved.

10. *Indicate in what manner the Lagrangian equations of motion*

$$\frac{d}{dt} \frac{dT}{d\xi'} - \frac{dT}{d\xi} = \frac{dV}{d\xi}, \quad \&c.$$

lead to the equations

$$A \frac{dp}{dt} + (C - B)qr = 0, \quad \&c.$$

for the motion of a solid body about a fixed point.

The expression of the *vis viva* function T is

$$T = \frac{1}{2}(Ap^2 + Bq^2 + Cr^2),$$

but this expression will not by itself lead to the equations of motion; we require to know also the expressions of p, q, r in terms of certain coordinates λ, μ, ν , which determine the position of the body in regard to axes fixed in space, and of the differential coefficients λ', μ', ν' of these coordinates in regard to the time; each of the quantities p, q, r will be a linear function of λ', μ', ν' ($p = a\lambda' + b\mu' + c\nu'$, &c.), containing in any manner whatever the coordinates λ, μ, ν . This being so, the equations of motion will be

$$\frac{d}{dt} \frac{dT}{d\lambda'} - \frac{dT}{d\lambda} = 0, \quad \&c.; \quad \frac{dT}{d\lambda'} = Ap \frac{dp}{d\lambda'} + Bq \frac{dq}{d\lambda'} + Cr \frac{dr}{d\lambda'},$$

where $\frac{dp}{d\lambda'}, \frac{dq}{d\lambda'}, \frac{dr}{d\lambda'}$ are each independent of λ', μ', ν' ; hence, in the equation, the only terms containing the differential coefficients of p, q, r , are the terms

$$\frac{dp}{d\lambda'} A \frac{dp}{dt} + \frac{dq}{d\lambda'} B \frac{dq}{dt} + \frac{dr}{d\lambda'} C \frac{dr}{dt}$$

of $\frac{dT}{d\lambda'}$; and hence, assuming that the equations of motion are the known equations

$$A \frac{dp}{dt} + (C - B)qr = 0,$$

it appears that the equation $\frac{d}{dt} \frac{dT}{d\lambda'} - \frac{dT}{d\lambda} = 0$ will assume the form

$$\begin{aligned} \frac{dp}{d\lambda'} \left\{ A \frac{dp}{dt} + (C - B)qr \right\} + \frac{dq}{d\lambda'} \left\{ B \frac{dq}{dt} + (A - C)rp \right\} \\ + \frac{dr}{d\lambda'} \left\{ C \frac{dr}{dt} + (B - A)pq \right\} = 0; \end{aligned}$$

there are of course two other equations only differing from this in that in place of λ' , they contain μ' and ν' respectively; and since p, q, r regarded as functions of λ', μ', ν' are independent functions, the determinant formed with the differential coefficients

p. 486 [534]

$$\frac{dp}{d\lambda'}, \quad \frac{dq}{d\lambda'}, \quad \frac{dr}{d\lambda'}, \quad \&c.$$

is not = 0; and the three equations are therefore equivalent (as they should be) to the equations

$$A \frac{dp}{dt} + (C - B)qr = 0, \quad \&c.$$

What precedes is a complete answer to the question, but in regard to the actual expressions of p, q, r , it may be remarked, that these quantities may be expressed very symmetrically in terms of the quantities

$$\lambda, \mu, \nu = \tan \frac{1}{2}\theta \cos f, \quad \tan \frac{1}{2}\theta \cos g, \quad \tan \frac{1}{2}\theta \cos h,$$

which determine the positions of the principal axes in regard to the axes fixed in space, by means of the angles of position $(\cos f, \cos g, \cos h)$ of the resultant axis, and the rotation θ about this axis; viz. writing $\kappa = 1 + \lambda^2 + \mu^2 + \nu^2$, we then have

$$\begin{aligned} \kappa p &= 2(\lambda' + \nu\mu' - \mu\nu'), \\ \kappa q &= 2(-\nu\lambda' + \mu' + \lambda\nu'), \\ \kappa r &= 2(\mu\lambda' - \lambda\mu' + \nu'), \end{aligned}$$

and the above result may be verified *a posteriori* without any difficulty. See *Camb. Math. Jour.*, vol. III. (1843), [6], p. 224, [*Coll. Math. Papers*, vol. I. p. 33].

11. Find in the Hamiltonian form,

$$\frac{d\eta}{dt} = \frac{dH}{d\varpi}, \quad \frac{d\varpi}{dt} = -\frac{dH}{d\eta}, \quad \&c.$$

the equations for the motion of a particle acted on by a central force.

Taking as coordinates r the radius vector, v the longitude, y the latitude, the equation of the *vis viva* function is

$$T = \frac{1}{2}\{r'^2 + r^2(\cos^2 y v'^2 + y'^2)\},$$

hence

$$\begin{aligned} \frac{dT}{dr'} &= r' = r \quad \text{suppose,} \\ \frac{dT}{dv'} &= r^2 \cos^2 y v' = v, \quad \frac{dT}{dy'} = r^2 y' = y, \end{aligned}$$

and the expression of T in terms of r, v, y , and of the new coordinates r, v, y is

$$T = \frac{1}{2} \left(r^2 + \frac{v^2}{r^2 \cos^2 y} + \frac{y^2}{r^2} \right);$$

whence writing

$$H = \frac{1}{2} \left(r^2 + \frac{v^2}{r^2 \cos^2 y} + \frac{y^2}{r^2} \right) - V,$$

p. 487 [534]

the equations are

$$\begin{aligned}\frac{dH}{dr} &= \frac{dr}{dt}, & \frac{dH}{dv} &= \frac{dv}{dt}, & \frac{dH}{dy} &= \frac{dy}{dt}, \\ \frac{dH}{dr} &= -\frac{dr}{dt}, & \frac{dH}{dv} &= -\frac{dv}{dt}, & \frac{dH}{dy} &= -\frac{dy}{dt}.\end{aligned}$$

We have

$$\begin{aligned}\frac{dH}{dr} &= r, & \frac{dH}{dv} &= \frac{v}{r^2 \cos^2 y}, & \frac{dH}{dy} &= \frac{y}{r^2}, \\ \frac{dH}{dr} &= -\frac{1}{r^3} \left(\frac{v^2}{\cos^2 y} + y^2 \right) - \frac{dV}{dr}, & \frac{dH}{dv} &= 0, & \frac{dH}{dy} &= \frac{v^2 \sin y}{r^2 \cos^3 y};\end{aligned}$$

and, substituting these values, the equations of motion present themselves as six equations of the first order between r, v, y, r, v, y , and t in the form

$$dt = \frac{dr}{r} = \frac{dv}{\frac{v}{r^2 \cos^2 y}} = \frac{dy}{\frac{y}{r^2}} = \frac{dr}{\frac{1}{r^3} \left(\frac{v^2}{\cos^2 y} + y^2 \right) + \frac{dV}{dr}} = \frac{dv}{0} = \frac{dy}{-\frac{v^2 \sin y}{r^2 \cos^3 y}}.$$

12. An unclosed polygon of $(m+1)$ vertices is constructed as follows: viz. the abscissae of the several vertices are $0, 1, 2, \dots, m$, and, corresponding to the abscissa k , the ordinate is equal to the chance of $(m+k)$ heads in $2m$ tosses of a coin; and m then continually increases up to any very large value: what information in regard to the successive polygons, and to the areas of any portions thereof, is afforded by the general results of the Theory of Probabilities?

It is somewhat more convenient to take account also of the abscissae $-1, -2, \dots, -m$, thereby obtaining a polygon of $2m+1$ vertices, symmetrical in regard to the axis of y . In such a polygon, the sum of the $2m+1$ ordinates is $= 1$; the central ordinate is the largest, and the ordinates continually diminish as k increases: moreover for any large value of m the area of the whole polygon is very nearly, and may be regarded as being, $= 1$; and the area between the ordinates corresponding to the abscissae $+k, -k$ as being equal to the probability of a number of heads between $m+k, m-k$, in the $2m$ tosses of the coin. A general result of the Theory of Probabilities is that in a great number of trials the several events tend to happen in the proportion of their respective probabilities; viz. in the case of the $2m$ tosses there is a tendency to an equal number of heads and tails. But observe that this does not mean that the probability of m heads and m tails increases with the number $2m$ of the trials; or even that, a being any given number, the probability of a number of heads between

p. 488 [534]

$m + a$ and $m - a$ increases with the number $2m$ of trials; on the contrary, it diminishes; what it does mean is that taking the limit of deviation to vary with m , say a number of heads between $m + \alpha m, m - \alpha m$, the probability of such a number increases with m ; viz. that taking α a fraction however small, m can be taken so large that the probability of a number of heads between $m + \alpha m, m - \alpha m$ in the $2m$ trials, shall be as nearly as we please = 1.

The conclusion in regard to the areas of the polygons is that, taking k any given value whatever, however large, the ratio (m being of course $> k$) which the area between the ordinates to the abscissae $m + k, m - k$ bears to the area of the whole polygon (or to unity) continually decreases as $2m$ increases, and ultimately vanishes; but contrariwise, taking α any given fraction whatever, however small, the ratio which the area between the ordinates to the abscissae $m + \alpha m, m - \alpha m$ bears to the area of the whole polygon (or to unity) continually increases as $2m$ increases, and ultimately becomes = 1.

13. Show that for the quadric cones which pass through six given points the locus of the vertices is a quartic surface having upon it twenty-five right lines; and, thence or otherwise, that for the quadric cones passing through seven given points the locus of the vertices is a sextic curve.

Suppose $U = 0, V = 0, W = 0, S = 0$ are any particular four quadric surfaces passing through the six points, say

$$U = (a, \dots)(x, y, z, w)^2, \quad V = (b, \dots)(x, y, z, w)^2, \quad \&c.;$$

then the equation of the general quadric surface through the six points will be

$$\alpha U + \beta V + \gamma W + \delta S = 0,$$

and this surface will be a cone, having (x, y, z, w) for the coordinates of its vertex, if only we have simultaneously

$$\alpha \frac{dU}{dx} + \beta \frac{dV}{dx} + \gamma \frac{dW}{dx} + \delta \frac{dS}{dx} = 0,$$

$$\&c. = 0,$$

$$\alpha \frac{dU}{dz} + \beta \frac{dV}{dz} + \gamma \frac{dW}{dz} + \delta \frac{dS}{dz} = 0,$$

$$\alpha \frac{dU}{dw} + \beta \frac{dV}{dw} + \gamma \frac{dW}{dw} + \delta \frac{dS}{dw} = 0.$$

Eliminating $(\alpha, \beta, \gamma, \delta)$ we have an equation $\nabla = 0$, where ∇ is the Jacobian or functional determinant

$$\frac{d(U, V, W, S)}{d(x, y, z, w)}$$

formed with the differential coefficients of the four functions (U, V, W, S) : the locus of the vertex is thus a quartic surface.

Calling the six points 1, 2, 3, 4, 5, 6, then taking as vertex any point in the line 12, the lines from such point to the points 1 and 2 coincide with the line 12, and we can through this line and the lines to the remaining points 3, 4, 5, 6 describe a quadric cone; the quartic surface therefore passes through the line 12; and similarly it passes through each of the fifteen lines 12, 13, ..., 56.

Again, taking the vertex anywhere in the line of intersection of the planes 123 and 456, we have an improper quadric cone, viz. the plane-pair formed by these two

p. 489 [534]

planes; the line in question is therefore a line of the quartic surface; and similarly the quartic surface contains each of the ten lines 123.456, 124.356, ..., 156.234. We have thus in all 25 lines on the quartic surface.

In the case of seven points 1, 2, 3, 4, 5, 6, 7, the locus is the curve of intersection of the quartic surfaces which correspond to the points 1, 2, 3, 4, 5, 6 and the points 1, 2, 3, 4, 5, 7 respectively: these have in common the ten lines 12, 13, 14, 15, 23, 24, 25, 34, 35, 45 (which it is easy to see do not form part of the required locus), and they have therefore, as a residual intersection, a curve of the order $16 - 10 = 6$, or sextic curve, which is the locus of the vertices of the cones which pass through the seven given points.

14. Show that the envelope of a variable circle having its centre on a given conic and cutting at right angles a given circle is a bicircular quartic; which, when the given conic and circle have double contact, becomes a pair of circles; and, by means of the last-mentioned particular case of the theorem, connect together the porisms arising out of the two problems:

(1) given two conics, to find a polygon of n sides inscribed in the one and circumscribed about the other;

(2) given two circles, to find a closed series of n circles each touching the two given circles and the two adjacent circles of the series.

The equation of the given circle is taken to be

$$(x - \alpha)^2 + (y - \beta)^2 = \gamma^2,$$

and that of the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. This being so, we have $a \cos \theta, b \sin \theta$ as the coordinates of a point on the conic, which point may be taken to be the centre of the variable circle, and introducing the condition that the two circles cut at right angles, the equation of the variable circle is

$$(x - a \cos \theta)^2 + (y - b \sin \theta)^2 = (\alpha - a \cos \theta)^2 + (\beta - b \sin \theta)^2 - \gamma^2,$$

that is,

$$x^2 + y^2 - \alpha^2 - \beta^2 + \gamma^2 - 2ax \cos \theta - 2by \sin \theta = 0,$$

where θ is the variable parameter; and the equation of the envelope therefore is

$$(x^2 + y^2 - \alpha^2 - \beta^2 + \gamma^2)^2 - 4a^2x^2 - 4b^2y^2 = 0,$$

which is a quartic curve; and writing herein $\frac{x}{z}, \frac{y}{z}$ in place of x, y the equation would be of the second order in regard to $x^2 + y^2, z$, and it thus appears that the curve has double points at each of the points $x^2 + y^2 = 0, z = 0$, viz. that the envelope is a bicircular quartic.

If the fixed circle touches the conic, then by a consideration of the figure it at once appears that the point of contact is a double point on the curve; and so if there is a double contact, then each of the points of contact is a double point on the curve. But in this case the curve is a bicircular quartic with four double points; viz. it is a pair of circles.

p. 490 [534]

The points in regard to the two conics is that, in general it is not possible to find any polygon of n sides satisfying the conditions; but that the case is also such that there exists an infinity of polygons; viz. any point whatever of the one conic may then be taken as a vertex of the polygon, and there are corresponding the the $(n+1)^{\text{th}}$ vertex still coincide with the first vertex, and there will be a polygon of n sides.

Now imagine that the conic touched by the sides is a circle having double contact with the other conic. Describe any one of the polygons, and with each vertex as centre describe the orthotomic circle, which will, it is clear, be a circle passing through the points of contact with the circles, each touching the two adjacent circles of the series. And by considering any other polygon, we have a like series of a circles; and by what precedes the envelope of all the circles of the several series is a pair of circles; that is, the circles of every series touch these two circles. We have consequently two circles, such that there exists an infinity of closed series of n circles, each circle touching the two fixed circles, and also the two adjacent circles of the series; which is the porism arising out of the second problem.

[535]

Note on the Problem of Envelopes.

[From the Messenger of Mathematics, vol. I. (1872), pp. 3, 4.]

THERE is a mode of looking at the problem of Envelopes, which, so far as I am aware, has not been explicitly noticed. Let $U = (x, y, z)^m$ be a function of the coordinates (x, y, z) ($\Theta = 0$ or (x, y, z) of θ , a function of the new set of coordinates (x, y, z) and (x', y', z') , θ being understood that when we write $\Theta = 0$ this expresses the existence of a relation between the coordinates (x, y, z) and the new set of coordinates (x', y', z') . Suppose that we have $\Theta = 0$, $U = 0$; that is, then the equation expresses θ as a function of (x, y, z) , (x', y', z') ; this is then a curve: the equation expressed values of parameters in coordinates (x, y, z) of a point P on the curve $U = 0$; and we may say that θ is the envelope (equivalent to four equations only)

$$U = 0, \quad \Theta = 0,$$

$$d_x U + \lambda d_x \Theta = 0, \quad d_y U + \lambda d_y \Theta = 0, \quad d_z U + \lambda d_z \Theta = 0,$$

and a particular integral $ay - b = 0$: the complete integral is in fact

$$(x - cy)^2 - 4r^2b^2 + 2(p - q)y^2 - rp^2 = 0,$$

satisfied, for $C = 0$, by $xy - n = 0$.

But, observe that the required envelope is the locus of the points of intersection of the curve $U = 0$, by curves in a particular point (x', y', z') of the other curve, $\Theta = 0$. So the curve $U = 0$ which belongs to a consecutive point (x', y', z') of the curve $U = 0$, considering therein (x', y', z') as the variable; by such a process the envelope of the curve y in question is a mere definition of what it does mean, and comes to be a fluxion. y then only chosen the values of A are $b, b', b'', \&c.$, and these values are then determined by the equations:— viz. if P is a point on the conic (x', y', z') , then conics, conditions (x', y', z') are conics on the same conic (x', y', z') as their values: and the curve in the plane which corresponds spend to the generating loop of the model.

p. 492 [535]

$\Theta = 0$, $U = 0$ to touch each other, we have a relation in (x', y', z') which is the equation of the point intersection of the curves $\Theta = 0$ belonging to the two consecutive values of the curve $U = 0$; that is, the equation of the required envelope is obtained as the condition that the curves $U = 0$, $\Theta = 0$ shall touch each other. But when the curves touch each, they have at the point of contact two consecutive points proportional, or we have simultaneously

$$U = 0, \quad \Theta = 0,$$

$$d_x U + \lambda d_x \Theta = 0,$$

$$d_y U + \lambda d_y \Theta = 0,$$

$$d_z U + \lambda d_z \Theta = 0,$$

the same equations as before, since $\Theta = 0$ and $\Theta = 0$ are the same function.

It is to be added that, when $a = u$, the equations

$$d_x U + \lambda d_x \Theta = 0,$$

$$d_y U + \lambda d_y \Theta = 0,$$

$$d_z U + \lambda d_z \Theta = 0,$$

are homogeneous in (x, y, z) , and we may by the elimination of (x, y, z) from these equations obtain an equation $\text{Disc. } (U \pm \lambda U) = 0$, say for shortness $\Delta = 0$, involving λ and also the coordinates (x', y', z') . Now it is a known theorem that the condition for the contact of the two curves $U = 0$, $U = 0$ may be obtained by expressing that the equation $\Delta = 0$ shall have a pair of equal roots, or, what is the same thing, by equating to the equation of the envelope of the curve $U = 0$, viz. by being so as shown the equation of the envelope of the curves $U = 0$, is in fact

$$\text{Disc. } \text{Disc.}_{(\lambda, 1)}(U + \lambda U) = 0;$$

viz. we first take the discriminant of the function $\Theta + \lambda U$ in regard to the coordinates (x, y, z) and then taking the discriminant in regard to λ of the function so obtained, b is zero. This is in many cases a more simple process than that of the direct elimination of x, y, z, λ from the equations.

[536]

Note on Lagrange's Demonstration of Taylor's Theorem.*[From the Messenger of Mathematics, vol. 1. (1872), pp. 22-24.]*

I TAKE the occasion of the publication of the last edition of Mr Todhunter's *Treatise on the Differential Calculus* to make some remarks on the demonstration in question. Mr Todhunter proposes to himself to establish a comprehensive view of the *Differential Calculus* as the method of Limits; but he very properly introduces in some cases demonstration founded upon other views of the subject, pointing out that this is the case, and explaining or justifying his objections. Thus (Chapter VI.) upon Taylor's Theorem, he remarks: "Before we offer a strict demonstration of the theorem in question, we shall notice the method which it was usual to adopt in treatises on the Differential Calculus not based on the doctrine of limits;" and then, after giving a demonstration depending on the relation $\frac{d}{dx}f(x+h) = \frac{d}{dh}f(x+h)$ he goes on to say "There are numerous objections to the method of the preceding article, and especially the use of an infinite series, without ascertaining that it is convergent, is inadmissible; we proceed then to prove Taylor's Theorem after the manner adopted by Mr Homersham Cox(*) in a demonstration of the equation

$$f(x+h) = f(x) + hf'(x) + \cdots + \frac{h^n}{[n+1]^{n+1}} f^{(n+1)}(x+\theta h),$$

(θ between 0 and 1) Taking "if the function $f^{(n)}(x+\theta h)$ is such that by making a sufficiently great the term $\frac{h^n}{[n+1]^{n+1}} f^{(n+1)}(x+\theta h)$ can be made as small as we please, then by carrying on the series

$$f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \&c.$$

* This demonstration is similar in principle to Laprange's but I think it is preferable; do, the gives in passing in this rather than the same value whether n is changed once, ; it on it. VIII, p. 9 (1851).

p. 494 [536]

to as many terms as we please we obtain a result differing as little as we please from $f(x+h)$. Under these circumstances then we may assert the truth of Taylor's theorem."

I share Abel's horror of divergent series(*); and I maintain the validity of Lagrange's demonstration. When by any objection one can expand a function in a series, for instance the function $\frac{1}{1-x}$, by division

$$\frac{1}{1-x} = 1 + x + \frac{x^2}{1-x} = 1 + x + x^2 + \frac{x^3}{1-x}, \&c.$$

$$1 + x + x^2 + x^3 + \&c.,$$

in the series $1 + x + x^2 + \&c.$, and write accordingly

$$\frac{1}{1-x} = 1 + x + x^2 + \&c.$$

all that is for ought be may mean is that the algebraical operations contained as far as we please will give the series of terms $1, x, x^2, \&c.$, or say the series of coefficients $1, 1, 1, \dots$. And of course with this meaning of the equation, the objection "we cannot that the series is convergent" would be wholly irrelevant, we do not say that in is, we do not care whether it is $\&c.$ or not. In further illustration, remark that we frequently use such an equation merely as the means of expressing the law of a series of numbers $a_0, a_1, a_2, \&c., a_n = a + b\theta, \&c.$; we make the assertion to be a definite process expressible in the form $a_n = \text{co. } x^n$ in $a_n = a_0 + a_1x + a_2x^2 + \&c.$ in question. Any objection that the series is not convergent would be simply irrelevant. Now an that the series shall be convergent is wholly employed for the expansion of $f(x+h)$ in regard to $f(x+h)$. It is impossible, in a quantitative algebra such as is presupposed in the method of limits, to put any meaning on the formula

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \&c.,$$

viz. $f(x+h)$ requires the same value $f(x+h)$ whether we change therein a into $x+h$ or h into $x+h$; and the expression on the right-hand side is the only series in h possessed of the same property. It is to be remarked that the equation contains in itself the definition of the operation of derivation, viz. the equation being true, $f(x)$ can only denote the coefficient of h in the expansion of $f(x+h)$, and what

¹ Peut-on imaginer rien de plus horrible que de débiter

$$0 = 1^n - 2^n + 3^n - 4^n + \&c.,$$

n étant un nombre entier positif?—(Œuvres, t. II, p. 266; [Nouv. Éd., 1881, t. II, p. 257].)

p. 495 [536]

really is shown is that admitting such an operation to be possible in regard not only to $f(x)$, but to $f'(x)$, &c., then the coefficients $f'(x)$, $\frac{f''(x)}{2!}$, &c., are obtained from $f(x)$ by the successive repetition of this operation and by dividing by the proper numerical denominator.

By what precedes, any objection in regard to convergency, I regard as irrelevant; and if it is said that the above-mentioned single assumption is not granted, I would either ask “What is a Function?”—or I would contend myself with the hypothetical statement: “If $f(x)$ be such that $f(x+h)$ is expansible in apoly, then Taylor’s theorem.

In regard to the demonstration given by Mr Todhunter, it impliedly assumes that x and h are both real, and (although doubtless possible) it would be considerably more difficult to find an analogous demonstration of the formula involving $f^{(n+1)}(x+\theta h)$ in the case of x and h imaginary. But the formula with the term in question is not (see Mr Todhunter consider it as being) Taylor’s theorem, ; to obtain from it Taylor’s theorem, we require (in the foregoing point of view) the property that $\lim_{n \rightarrow \infty} f^{(n+1)}(x+\theta h)$ is exposable in a series involving h^{n+1} and the higher powers of h , that is, the very property that $f(x+h)$ is exposable in positive powers of h .

Moreover admitting that the formula with the term $f^{(n+1)}(x+\theta h)$ is demonstratable for imaginary values of x , h , then formula is unenchapsie in the case where $x+h$ is one or both a symbol or symbols of operation; whereas it would certainly have to include unmoral magnitude, and if it is considered as meaning anything, then the equation in question is mere definition of what it does mean, and ceases to be a theorem in regard to $f(x+h)$. It is impossible, in a quantitative algebra such as is presupposed in the method of limits, to put any meaning on the formula

$$f\left(\frac{d}{dx} + h\right) = f\left(\frac{d}{dx}\right) + hf'\left(\frac{d}{dx}\right) + \&c.,$$

which however I regard as a legitimate particular form of Taylor’s theorem.

[537]

Solutions of a Smith's Prize Paper for 1871.*[From the Messenger of Mathematics, vol. I. (1872), pp. 37-47, 71-77, 89-95.]*

1. *A point moves in a plane with a given velocity, and also with a given angular velocity about a fixed point in the plane: show that the locus is either a circle passing through the fixed point, or else a circle having the fixed point for its centre; and explain the relation between the two solutions.*

We have in general

$$v^2 = \left(\frac{dr}{dt}\right)^2 + r^2 \left(\frac{d\phi}{dt}\right)^2,$$

and in the present question, taking the fixed point as the origin, and measuring θ from any fixed line through this point,

$$\frac{d\theta}{dt} = a, \quad V^2 = \left(\frac{dr}{dt}\right)^2 + r^2 a^2,$$

where V, a are given constants. Hence

$$\left(\frac{dr}{dt}\right)^2 + a^2 r^2 = V^2, \quad \left(\frac{dr}{dt}\right)^2 = V^2 - a^2 r^2,$$

or, writing $V = aa$,

$$\left(\frac{dr}{dt}\right)^2 = a^2(a^2 - r^2);$$

therefore

$$d\theta = \frac{dr}{\sqrt{(a^2 - r^2)}},$$

or

$$\theta + \beta = \sin^{-1} \frac{r}{a}, \quad (\beta \text{ the constant of integration,})$$

p. 497 [537]

that is,

$$r = a \sin(\theta + \beta),$$

which is the equation of a circle (radius = $\frac{1}{2}a$) passing through the fixed point. In fact, the point moving in such a circle with a constant velocity, moves about the centre with a constant angular velocity; and about any fixed point in the circumference with an angular velocity which is one-half that about the centre, and is therefore also constant.

Treating θ as a variable parameter, to obtain the envelope we have

$$0 = a \cos(\theta + \beta),$$

that is, $\theta + \beta = \frac{\pi}{2}$, and therefore $r = a$, which is the equation of a circle (radius = a) having the fixed point for its centre. This is consequently the singular solution.

2. Determine the system of curves which satisfy the differential equation

$$dx(\sqrt{(1+x^2+xy+xy^2)}) + dy(\sqrt{(1+x^2+xy+xy^2)}) = 0;$$

and that the curve which passes through the point $x = 0, y = 0$, contains as a part of itself the conic

$$x^2 + y^2 + 2xy\sqrt{(1+x^2+xy+xy^2)} = 1.$$

The equation is integrable *per se*, viz. we have

$$x\sqrt{(1+x^2)} + \log\{x + \sqrt{(1+x^2)}\} + \frac{1}{2}y\sqrt{(1+y^2)} + \log\{y + \sqrt{(1+y^2)}\} = U$$

or, denoting the constant so that the differential equation has its general solution

$$x\sqrt{(1+x^2)} + y\sqrt{(1+y^2)} + 2xy = (1+x^2+xy+y^2) + \log \frac{\{x + \sqrt{(1+x^2)}\}\{y + \sqrt{(1+y^2)}\}}{1-C} = 0,$$

which is evidently a transcendental curve; it may however be shown that, if

$$x^2 + y^2 + 2xy\sqrt{(1+x^2+xy+xy^2)} - 1 = 0,$$

so that the equation of the curve is then made obvious, that is, the transcendental curve $y = axy + b$, which is the elementary curve, is in this manner included as a part of the curve.

[At a simple illustration as to how this may happen, take the transcendental curve $y = axy + b$, which is the elementary curve, in this manner included as itself.]

p. 498 [537]

We have, from the equation of the conic

$$\{x + y\sqrt{(1 + n^2)}\}^2 = n^2(1 + y^2),$$

that is,

$$x + y\sqrt{(1 + n^2)} = \pm n\sqrt{(1 + y^2)},$$

but considering the radicals as positive, the sign must be taken so that we have simultaneously $x = 0$, $y = n$. We have therefore

$$x + y\sqrt{(1 + n^2)} = n\sqrt{(1 + y^2)},$$

and similarly

$$y + x\sqrt{(1 + n^2)} = n\sqrt{(1 + x^2)}.$$

Then

$$\begin{aligned} n\{x\sqrt{(1 + x^2)} + y\sqrt{(1 + y^2)}\} &= 2xy + (x^2 + y^2)\sqrt{(1 + n^2)} \\ &= 2xy + \sqrt{(1 + n^2)}\{n^2 - 2xy\sqrt{(1 + n^2)}\} \\ &= n^2\{\sqrt{(1 + n^2)} - 2xy\}, \end{aligned}$$

which is the first of the relations in question; and

$$\begin{aligned} n^2\{x + \sqrt{(1 + x^2)}\}\{y + \sqrt{(1 + y^2)}\} &= n^2xy + nx\{x + y\sqrt{(1 + n^2)}\} + ny\{y + x\sqrt{(1 + n^2)}\} \\ &\quad + xy + (x^2 + y^2)\sqrt{(1 + n^2)} + xy(1 + n^2) \\ &= \{n + \sqrt{(1 + n^2)}\}\{x^2 + y^2 + 2xy\sqrt{(1 + n^2)}\} \\ &= \{n + \sqrt{(1 + n^2)}\}n^2, \end{aligned}$$

which is the second of the two relations. And the theorem is thus proved.

[The foregoing is the easiest and most obvious solution, but it is interesting to consider the question differently, as follows:

Write

$$Q = \frac{\{x + \sqrt{(1 + x^2)}\}\{y + \sqrt{(1 + y^2)}\}}{n + \sqrt{(1 + n^2)}},$$

we have

$$\begin{aligned} Q\{\sqrt{(n^2 + 1)} + n\} &= \{\sqrt{(1 + x^2)} + x\}\{\sqrt{(1 + y^2)} + y\} = A + B, \\ Q^{-1}\{\sqrt{(n^2 + 1)} - n\} &= \{\sqrt{(1 + x^2)} - x\}\{\sqrt{(1 + y^2)} - y\} = A - B, \end{aligned}$$

if

$$\begin{aligned} A &= \sqrt{(1 + x^2)}\sqrt{(1 + y^2)} + xy, \\ B &= x\sqrt{(1 + y^2)} + y\sqrt{(1 + x^2)}; \end{aligned}$$

and then

$$\begin{aligned} AB &= x^2y\sqrt{(1 + y^2)} + xy^2\sqrt{(1 + x^2)} \\ &\quad + y(1 + x^2)\sqrt{(1 + y^2)} + x(1 + y^2)\sqrt{(1 + x^2)} \\ &= x\sqrt{(1 + x^2)} + y\sqrt{(1 + y^2)} + 2xyB, \end{aligned}$$

p. 499 [537]

that is,

$$\begin{aligned} & Q^2\{2n^2 + 1 + 2n\sqrt{(1+n^2)}\} - \frac{1}{Q^2}\{2n^2 + 1 - 2n\sqrt{(1+n^2)}\} \\ &= 4\{x\sqrt{(1+x^2)} + y\sqrt{(1+y^2)}\} \\ &+ 4xy \left[Q\{\sqrt{(1+n^2)} + n\} - \frac{1}{Q}\{\sqrt{(1+n^2)} - n\} \right], \end{aligned}$$

whence

$$\begin{aligned} & 4\{x\sqrt{(1+x^2)} + y\sqrt{(1+y^2)} + 2nxy - n\sqrt{(1+n^2)}\} \\ &= Q^2\{2n^2 + 1 + 2n\sqrt{(1+n^2)}\} - \frac{1}{Q^2}\{2n^2 + 1 - 2n\sqrt{(1+n^2)}\} \\ &+ 8nxy - 4n\sqrt{(1+n^2)} \\ &- 4xy \left[Q\{\sqrt{(1+n^2)} + n\} - \frac{1}{Q}\{\sqrt{(1+n^2)} - n\} \right] \\ &= \left(Q^2 - \frac{1}{Q^2} \right) (2n^2 + 1) + \left(Q^2 + \frac{1}{Q^2} - 2 \right) 2n\sqrt{(1+n^2)} \\ &- \left(Q - \frac{1}{Q} \right) 4xy\sqrt{(1+n^2)} - 4 \left(Q + \frac{1}{Q} - 2 \right) nxy \\ &= (Q-1) \left\{ \frac{(Q+1)(Q^2+1)}{Q^2} (2n^2 + 1) + \frac{(Q-1)(Q+1)^2}{Q^2} 2n\sqrt{(1+n^2)} \right. \\ &\quad \left. - \frac{4(Q+1)}{Q} xy\sqrt{(1+n^2)} - 4 \frac{(Q-1)}{Q} nxy \right\} \\ &= (Q-1)\Omega \quad \text{suppose,} \end{aligned}$$

that is,

$$x\sqrt{(1+x^2)} + y\sqrt{(1+y^2)} + 2nxy - n\sqrt{(1+n^2)} = \frac{1}{4}(Q-1)\Omega.$$

And the integral equation is

$$\frac{1}{4}(Q-1)\Omega + \log Q = C,$$

which, for $C = 0$, is satisfied by $Q = 1$.

Now starting from

$$Q = \frac{\{x + \sqrt{(1+x^2)}\}\{y + \sqrt{(1+y^2)}\}}{n + \sqrt{(1+n^2)}},$$

we have

$$\begin{aligned} \sqrt{(1+x^2)} + x &= Q\{\sqrt{(1+n^2)} + n\}\{\sqrt{(1+y^2)} - y\}, \\ \sqrt{(1+x^2)} - x &= \frac{1}{Q}\{\sqrt{(1+n^2)} - n\}\{\sqrt{(1+y^2)} + y\}, \end{aligned}$$

p. 500 [537]

and thence

$$2x = K\sqrt{(1+y^2)} - Ly,$$

if

$$K = Q\{\sqrt{(1+n^2)} + n\} - \frac{1}{Q}\{\sqrt{(1+n^2)} - n\},$$

$$L = Q\{\sqrt{(1+n^2)} + n\} + \frac{1}{Q}\{\sqrt{(1+n^2)} - n\};$$

wherefore

$$L^2 - K^2 = 4.$$

Moreover

$$(2x + Ly)^2 = K^2(1 + y^2),$$

that is,

$$4x^2 + (L^2 - K^2)y^2 + 4Lxy = K^2;$$

or, what is the same thing,

$$x^2 + y^2 + Lxy = \frac{1}{4}(L^2 - 4),$$

which is the rationalised form of

$$Q = \frac{\{x + \sqrt{(1+x^2)}\}\{y + \sqrt{(1+y^2)}\}}{n + \sqrt{(1+n^2)}}.$$

And if $Q = 1$ then $L = 2\sqrt{(1+n^2)}$, $\frac{1}{4}(L^2 - 4) = n^2$, so that this equation is

$$x^2 + y^2 + 2xy\sqrt{(1+n^2)} - n^2 = 0;$$

or, when $C = 0$, the complete integral is satisfied by

$$\frac{\{x + \sqrt{(1+x^2)}\}\{y + \sqrt{(1+y^2)}\}}{n + \sqrt{(1+n^2)}} = 1,$$

that is, by

$$x^2 + y^2 + 2xy\sqrt{(1+n^2)} - n^2 = 0.$$

We may without difficulty rationalise, and present the result as follows: the equation

$$\begin{aligned} & \left\{ 2\left(x + \frac{1}{x}\right) + \left(n - \frac{1}{n}\right)\left(y - \frac{1}{y}\right) \right\} \left(1 + \frac{1}{x^2}\right) dx \\ & + \left\{ 2\left(y + \frac{1}{y}\right) + \left(n - \frac{1}{n}\right)\left(x - \frac{1}{x}\right) \right\} \left(1 + \frac{1}{y^2}\right) dy = 0, \end{aligned}$$

has the complete integral

$$x^2 - \frac{1}{x^2} + y^2 - \frac{1}{y^2} + \left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right)\left(n - \frac{1}{n}\right) - \left(n^2 - \frac{1}{n^2}\right) = C + 4 \log \frac{xy}{n},$$

and a particular integral $xy - n = 0$: the complete integral is in fact

$$(n - xy)\{-n^3x^2y^2 + n^2xy(-x^2 - y^2 + 1) + n(x^2y^2 - x^2 - y^2) - xy\} = x^2y^2n^2 \left(C + 4 \log \frac{xy}{n}\right),$$

satisfied, for $C = 0$, by $xy - n = 0$.]

p. 501 [537]

3. Write $\alpha = b - c$, $\beta = c - a$, $\gamma = a - b$; then considering the three circles and the three conics

$$\begin{aligned}(x - a)^2 + y^2 &= -\beta\gamma, & \frac{x^2}{bc} + \frac{y^2}{K + bc} &= 1, \\(x - b)^2 + y^2 &= -\gamma\alpha, & \frac{x^2}{ca} + \frac{y^2}{K + ca} &= 1, \\(x - c)^2 + y^2 &= -\alpha\beta, & \frac{x^2}{ab} + \frac{y^2}{K + ab} &= 1,\end{aligned}$$

where K is arbitrary; it is required to show that if a variable circle having its centre on one of the conics cuts at right angles the corresponding circle, the envelope is in each of the three cases one and the same bicircular quartic.

Consider the circle $(x - a)^2 + y^2 = -\beta\gamma$ and the conic

$$\frac{x^2}{bc} + \frac{y^2}{K + bc} = 1;$$

the coordinates of a point on the conic are $\cos \theta \sqrt{bc}$, $\sin \theta \sqrt{K + bc}$, where θ is a variable parameter; say for a moment these values are p and q . The equation of the variable circle is

$$(x - p)^2 + (y - q)^2 = r^2,$$

and in order that this may cut at right angles the circle

$$(x - a)^2 + y^2 = -\beta\gamma,$$

we must have

$$(p - a)^2 + q^2 = r^2 - \beta\gamma,$$

or, substituting for r^2 its value from this equation, the equation of the variable circle is

$$(x - p)^2 + (y - q)^2 = (a - p)^2 + q^2 + \beta\gamma,$$

that is,

$$x^2 + y^2 - a^2 - \beta\gamma - 2p(x - a) - 2qy = 0,$$

viz. this is

$$(x^2 + y^2 - a^2 - \beta\gamma) - 2(x - a)\sqrt{bc}\cos \theta - 2y\sqrt{K + bc}\sin \theta = 0.$$

Hence taking the envelope in regard to θ , the equation is

$$(x^2 + y^2 - a^2 - \beta\gamma)^2 - 4(x - a)^2bc - 4y^2(K + bc) = 0,$$

that is,

$$(x^2 + y^2 - ab - ac + bc)^2 - 4(x - a)^2bc - 4y^2(K + bc) = 0,$$

or, what is the same thing,

$$\begin{aligned}(x^2 + y^2)^2 - 2(bc + ca + ab)(x^2 + y^2) - 4Ky^2 + 8abcx \\ + b^2c^2 + c^2a^2 + a^2b^2 - 2a^2bc - 2b^2ca - 2c^2ab = 0,\end{aligned}$$

viz. this equation, being symmetrical in regard to a, b, c , is the same equation as would have been obtained from either of the other conics and the corresponding circle; and from the form of the equation it is clear that the curve is a bicircular quartic.

4. Show that the caustic by refraction for parallel rays of a circle, radius c , index of refraction μ , is the same curve as the caustic by refraction for parallel rays of the concentric circle, radius $\frac{c}{\mu}$, index of refraction $\frac{1}{\mu}$.

Take as usual $\mu > 1$. Imagine the ray AP (fig. 1) parallel to the axis of x , incident at P on the circle radius c , and let the refracted ray after cutting the circle radius $\frac{c}{\mu}$, cut it again in Q , and then cut the axis in R . Take ϕ, ϕ' for the angles of incidence and refraction; $\sin \phi = \mu \sin \phi'$.

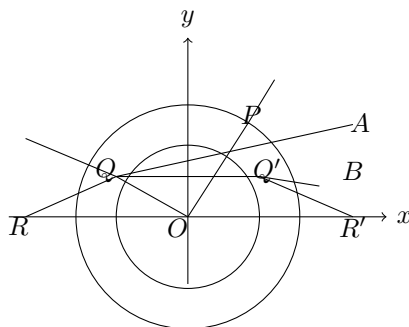


Fig. 1.

Moreover in the triangle PQO , we have $\sin Q : \sin P = c : \frac{c}{\mu}$; that is, $\sin Q = \mu \sin P = \mu \sin \phi' = \sin \phi$; or $\angle Q = \phi$. And then in the triangle RQO , $\angle R = \phi - \phi'$, $\angle Q = 180^\circ - \phi$, whence $\angle O = \phi'$, that is, $\angle QOR = \phi'$.

Consider now a ray BQ incident at Q and refracted in the direction QR ; the index of refraction being $\frac{1}{\mu}$, that is, the denser medium being on the outside of the small circle. Taking θ, θ' for the angles of incidence and refraction, we have $\sin \theta = \frac{1}{\mu} \sin \theta'$; but, the refracted ray being by hypothesis QR , we have by what precedes $\theta' = \phi$, hence $\sin \theta = \frac{1}{\mu} \sin \phi = \sin \phi'$, that is, $\theta = \phi'$, or $\angle BQO = \angle QOR$, that is, the incident ray BQ is parallel to the axis of x . We have thus two pencils of rays each parallel to the axis, such that for any ray AP of the first pencil there is a corresponding ray BQ of the second pencil, the rays AP and BQ each giving rise to the same refracted ray PQR ; hence the two pencils have the same caustic.

[It is proper to remark that for the ray BQ it has been assumed that the refraction takes place not at Q' where it first meets the small circle, but at Q ; if

we consider the refraction at Q' , then the index of refraction is still to be $= \frac{1}{\mu}$, that is, the denser medium must now be inside the small circle; the refracted ray is in the direction $R'Q'$ situate symmetrically with RQ on the opposite side of the axis of y ; and it would at first sight appear that the caustic was a curve equal and similar to the original caustic, but situate on the opposite side of the axis of y . But geometrically the complete caustic consists of two equal and similar portions situate on opposite sides of the axis of y ; so that we really obtain, not an equal and opposite caustic, but in each case one and the same caustic.

I originally obtained the theorem in a different manner; viz. the equation for the caustic for the first pencil of rays was found to be

$$(1 - \mu^2) \frac{x}{c} = \left\{ 1 - \mu^{\frac{4}{3}} \left(\frac{y}{c} \right)^{\frac{2}{3}} \right\}^{\frac{3}{2}} + \mu \left\{ 1 - \mu^{-\frac{2}{3}} \left(\frac{y}{c} \right)^{\frac{2}{3}} \right\}^{\frac{3}{2}},$$

which equation (as is easily seen) remains unaltered when c, μ are changed into $\frac{c}{\mu}, \frac{1}{\mu}$ respectively.— See my “Memoir on Caustics,” *Phil. Trans.*, t. CXLVII. (1857), [145], p. 285.]

5. *Given at each point of space the direction-cosines (α, β, γ) of a line through that point: it is required to find the conditions in order that the lines may be not a triple but a double system.*

For any given point P the values of the quantities α, β, γ which determine the direction of the line through that point are given as functions of the coordinates (x, y, z) of the point P . Hence passing from a point P to a consecutive point P' on the line, the coordinates of P' will be $x + \rho\alpha, y + \rho\beta, z + \rho\gamma$; and the values of α, β, γ for the point P' will be

$$\begin{aligned} \alpha + \rho \left(\alpha \frac{d\alpha}{dx} + \beta \frac{d\alpha}{dy} + \gamma \frac{d\alpha}{dz} \right), \\ \beta + \rho \left(\alpha \frac{d\beta}{dx} + \beta \frac{d\beta}{dy} + \gamma \frac{d\beta}{dz} \right), \\ \gamma + \rho \left(\alpha \frac{d\gamma}{dx} + \beta \frac{d\gamma}{dy} + \gamma \frac{d\gamma}{dz} \right). \end{aligned}$$

But if the lines form a double system, we must have the same line for the point P , and for any other point P' on the line; and in particular the same line for the point P , and for the consecutive point P' . Hence as conditions for the double system we obtain

$$\begin{aligned} \alpha \frac{d\alpha}{dx} + \beta \frac{d\alpha}{dy} + \gamma \frac{d\alpha}{dz} &= 0, \\ \alpha \frac{d\beta}{dx} + \beta \frac{d\beta}{dy} + \gamma \frac{d\beta}{dz} &= 0, \\ \alpha \frac{d\gamma}{dx} + \beta \frac{d\gamma}{dy} + \gamma \frac{d\gamma}{dz} &= 0. \end{aligned}$$

But in virtue of the relation $\alpha^2 + \beta^2 + \gamma^2 = 1$, we have

$$\begin{aligned}\alpha \frac{d\alpha}{dx} + \beta \frac{d\beta}{dx} + \gamma \frac{d\gamma}{dx} &= 0, \\ \alpha \frac{d\alpha}{dy} + \beta \frac{d\beta}{dy} + \gamma \frac{d\gamma}{dy} &= 0, \\ \alpha \frac{d\alpha}{dz} + \beta \frac{d\beta}{dz} + \gamma \frac{d\gamma}{dz} &= 0.\end{aligned}$$

Hence subtracting the corresponding equations we have three equations, which are at once seen to be equivalent to the two equations

$$\frac{d\beta}{dz} - \frac{d\gamma}{dy} : \frac{d\gamma}{dx} - \frac{d\alpha}{dz} : \frac{d\alpha}{dy} - \frac{d\beta}{dx} = \alpha : \beta : \gamma,$$

equations which must be satisfied identically, whatever are the values of (x, y, z) . The equations have been obtained as necessary conditions; they are, in fact, the sufficient conditions for a double system; for the line being unaltered in passing from P to P' , it remains unaltered when we pass to the following point P'' , and so on; that is, for the passage to any point Q whatever on the line.

COR. If the equation $\alpha dx + \beta dy + \gamma dz = 0$ be integrable by a factor, it must be integrable *per se*: in fact, the condition that it may be integrable by a factor is

$$\alpha \left(\frac{d\beta}{dz} - \frac{d\gamma}{dy} \right) + \beta \left(\frac{d\gamma}{dx} - \frac{d\alpha}{dz} \right) + \gamma \left(\frac{d\alpha}{dy} - \frac{d\beta}{dx} \right) = 0.$$

But we have

$$\frac{d\beta}{dz} - \frac{d\gamma}{dy} = k\alpha, \quad \&c.,$$

and the equation thus becomes

$$k(\alpha^2 + \beta^2 + \gamma^2) = 0,$$

that is, $k = 0$, and therefore

$$\frac{d\beta}{dz} - \frac{d\gamma}{dy} = 0, \quad \frac{d\gamma}{dx} - \frac{d\alpha}{dz} = 0, \quad \frac{d\alpha}{dy} - \frac{d\beta}{dx} = 0.$$

Hence, also, if the lines cut at right angles a surface, we must have $\alpha dx + \beta dy + \gamma dz$ a complete differential.

The foregoing theory is given in Sir W. R. Hamilton's "Memoir on Ray-Systems."

6. If $X = 0, Y = 0, Z = 0, W = 0$ are four given conies in the same plane and having a common point: show that, in the system of conies $aX + bY + cZ + dW = 0$, there are in general four (improper) conies the equations of which may be taken to be $x^2 = 0, y^2 = 0, xz = 0, yz = 0$.

Taking the conies to pass through the point $x = 0, y = 0$; their equations will be of the form

$$\begin{aligned} X &= a_1x^2 + 2h_1xy + b_1y^2 + 2f_1yz + 2g_1zx = 0, \\ Y &= a_2x^2 + 2h_2xy + b_2y^2 + 2f_2yz + 2g_2zx = 0, \\ Z &= a_3x^2 + 2h_3xy + b_3y^2 + 2f_3yz + 2g_3zx = 0, \\ W &= a_4x^2 + 2h_4xy + b_4y^2 + 2f_4yz + 2g_4zx = 0. \end{aligned}$$

Now multiplying by the indeterminate quantities $\alpha, \beta, \gamma, \delta$, the three ratios $\alpha : \beta : \gamma : \delta$ may be determined so that the terms in yz, zx shall vanish, and the terms in x^2, xy, y^2 be a perfect square: we thus arrive at a quadric equation for any one of the ratios, say $\alpha : \beta$, the remaining ratios being then linearly determined; viz. there are two sets of values of $\alpha, \beta, \gamma, \delta$: and changing the coordinates (x, y) , the two resulting forms may be represented by $x^2 = 0, y^2 = 0$.

And it is clear that we thus have in the system of conies $\alpha X + \beta Y + \gamma Z + \delta W = 0$, four conies the equations of which may be represented by

$$\begin{aligned} X' &= x^2 = 0, \\ Y' &= y^2 = 0, \\ Z' &= h_3xy + f_3yz + g_3zx = 0, \\ W' &= h_4xy + f_4yz + g_4zx = 0, \end{aligned}$$

where of course the coefficients f, g, h have new values.

We may then form the equations

$$\begin{aligned} \alpha X' + f_4 Z' - f_3 W' &= x\{\alpha x + (f_4 h_3 - f_3 h_4)y + (f_4 g_3 - f_3 g_4)z\}, \\ \beta Y' - g_4 Z' + g_3 W' &= y\{(g_3 h_4 - g_4 h_3)x + \beta y + (f_4 g_3 - f_3 g_4)z\}, \end{aligned}$$

so that, by writing $\alpha = g_3 h_4 - g_4 h_3$ and $\beta = f_4 h_3 - f_3 h_4$, the terms in $\{ \}$ will be one and the same linear function of (x, y, z) ; that is, changing the z so as to denote the linear function in question by z , we have as conies of the series $xz = 0$, and $yz = 0$, that is, we have in the series the four conies $x^2 = 0, y^2 = 0, xz = 0, yz = 0$; whence also any other conic of the series, and consequently each of the original four conies, may be represented by an equation of the form

$$ax^2 + by^2 + 2fyz + 2gzx = 0.$$

7. The coordinates (x, y, z, w) of a point P in space are connected with the coordinates (x', y', z') of a point P' in a plane by the equations

$$x : y : z : w = X' : Y' : Z' : W',$$

where X', Y', Z', W' are quadric functions of (x', y', z') such that $X' = 0, Y' = 0, Z' = 0, W' = 0$ represent conies having a common point: show that the locus of P is a cubic scroll (skew surface of the third order): and find the curves in the plane which correspond to the generating lines of the scroll.

The equations $x : y : z : w = X' : Y' : Z' : W'$ are three equations containing the indeterminate parameters $x' : z'$ and $y' : z'$, so that eliminating these we have between (x, y, z, w) a single (homogeneous) equation representing a surface. To each point (x', y', z') of the plane, there corresponds a single point of the surface, and to each point (x, y, z, w) of the surface a single point (x', y', z') of the plane. The only exception is that for the common point of the four conies, the ratios $x : y : z : w$ are essentially indeterminate, and there is not corresponding hereto any determinate point of the surface.

To find the order of the surface, consider its intersection with any arbitrary line

$$ax + by + cz + dw = 0, \quad a_1x + b_1y + c_1z + d_1w = 0.$$

We have corresponding hereto in the plane the points of intersection of the conies

$$aX' + bY' + cZ' + dW' = 0, \quad a_1X' + b_1Y' + c_1Z' + d_1W' = 0,$$

viz. these are conies each of them passing through the common point of the four conies, and therefore they intersect besides in three points: that is, the order of the surface is = 3.

To show that the common point ought to be (as above) excluded, some further explanation is desirable. To the section of the surface by the plane $ax + by + cz + dw = 0$, corresponds the conic $aX' + bY' + cZ' + dW' = 0$; and similarly to the section by the plane $a_1x + b_1y + c_1z + d_1w = 0$, corresponds the conic $a_1X' + b_1Y' + c_1Z' + d_1W' = 0$. Now to the common point considered as belonging to the first conic there corresponds a determinate point of the surface; and to the common point considered as belonging to the second conic there corresponds a determinate point of the surface; but these are two distinct points on the surface: so that corresponding to the common point of the four conies, there is not on the surface any point of intersection of the two plane sections; but these intersect in only three points of the surface; viz. the line of intersection of the two planes meets the surface in three points: or the surface is a cubic surface.

The same result may be obtained, and it may be further shown that the surface is a scroll, by means of the property in the foregoing question 6; viz. it thereby appears that each of the functions X', Y', Z', W' may be taken to be of the form $ax'^2 + by'^2 + fy'z' + gz'x'$; hence replacing the original coordinates x, y, z, w , by properly selected linear functions of these coordinates, the given relations may be presented in the form

$$x : y : z : w = x'^2 : y'^2 : x'z' : y'z',$$

whence eliminating, we have

$$xw^2 - yz^2 = 0$$

the equation of a cubic scroll, having the line $z = 0, w = 0$ for a double line, and the line $x = 0, y = 0$ for a directrix line. The equations of a generating line of the scroll

are, it is clear, $z - \theta w = 0$, $x - \theta^2 y = 0$, where θ is a variable parameter; and corresponding hereto in the plane we have the line $x' - \theta y' = 0$, viz. this is any line through the common intersection of the four conies.

8. If U, V are binary functions of the form $(a, b, \dots)(x, y)^m$ with arbitrary coefficients, and if the equations $U = 0$, $V = 0$ have a common root, show how this can be determined in terms of the derived functions of the Resultant in regard to the coefficients of either function.

Show what results in regard to the common root can be obtained when the coefficients are not all of them arbitrary but (1) each or either of the functions depends in any manner whatever on a set of arbitrary coefficients not entering into the other function, (2) the two functions depend in any manner whatever on one and the same set of arbitrary coefficients.

How is the theory modified when, instead of the two equations, there is a single equation $U = 0$ having a double root?

Suppose

$$U = (a, b, \dots)(x, y)^m \left(= ax^m + \frac{m}{1}bx^{m-1}y + \&c. \right),$$

$$V = (a', b', \dots)(x, y)^{m'} \left(= a'x^{m'} + \frac{m'}{1}b'x^{m'-1}y + \&c. \right).$$

Then if R is the resultant, the equation $R = 0$ is the relation which must exist between the coefficients $(a, b, \dots)$ and $(a', b', \dots)$ in order that the equations $U = 0$ and $V = 0$ may have a common root (that is, in order that the functions U, V may have a common factor $x - ay$). Imagine the relation subsisting, and that x, y are the values belonging to the common root, or (what is the same thing) that we have $x - ay = 0$; we have then simultaneously $U = 0$, $V = 0$, $R = 0$. Now suppose the coefficients $a, b, \dots$ to be infinitesimally varied in such manner that U, V have still a common root; say the new values are $a + \delta a$, $b + \delta b, \dots$: this implies between $\delta a, \delta b, \dots$ the relation

$$\frac{dR}{da}\delta a + \frac{dR}{db}\delta b + \dots = 0.$$

But the common factor $x - ay$ is a factor of the unaltered equation $V = 0$; and the values of (x, y) are thus unaltered, viz. the equation $U = 0$ is satisfied with the original values of (x, y) ; so that we have

$$\frac{dU}{da}\delta a + \frac{dU}{db}\delta b + \dots = 0,$$

or, what is the same thing,

$$x^m\delta a + mx^{m-1}y\delta b + \dots = 0,$$

an equation which must agree with the former one, that is, we have

$$x^m : mx^{m-1}y : \&c. = \frac{dR}{da} : \frac{dR}{db} : \&c.,$$

a series of equations giving the value of the common root $\frac{x}{y}(=a)$ in the several forms

$$\frac{1}{m} \frac{x}{y} = \frac{dR}{da} \div \frac{dR}{db}, \quad \frac{2}{m-1} \frac{x}{y} = \frac{dR}{db} \div \frac{dR}{dc}, \quad \&c.$$

And it is clear that we have in like manner

$$x^{m'} : m' x^{m'-1} y : \&c. = \frac{dR}{da'} : \frac{dR}{db'} : \&c..$$

It is clear that if U involves, in any manner whatever, the coefficients $a, b, \dots$ which do not enter into the function V , then we have in precisely the same manner

$$\frac{dU}{da} : \frac{dU}{db} : \&c. = \frac{dR}{da} : \frac{dR}{db} : \&c.,$$

a system of equations satisfied by the values x, y which belong to the common root.

But if the coefficients $a, b, \dots$ are contained in any manner whatever in both of the functions U, V ; then by altering $a, b, \dots$ we alter the common root; say that $x + \delta x, y + \delta y$ belong to its new value; then we have

$$\begin{aligned} \frac{dU}{dx} \delta x + \frac{dU}{dy} \delta y + \frac{dU}{da} \delta a + \frac{dU}{db} \delta b + \dots &= 0, \\ \frac{dV}{dx} \delta x + \frac{dV}{dy} \delta y + \frac{dV}{da} \delta a + \frac{dV}{db} \delta b + \dots &= 0. \end{aligned}$$

Now the values of x, y which satisfy $U = 0, V = 0$ also satisfy

$$\frac{dU}{dx} \frac{dV}{dy} - \frac{dU}{dy} \frac{dV}{dx} = 0;$$

hence from the foregoing two equations eliminating δx or δy , the other of these two quantities will disappear of itself, and we thus obtain an equation

$$A \delta a + B \delta b + \dots = 0,$$

which must agree with the above equation

$$\frac{dR}{da} \delta a + \frac{dR}{db} \delta b + \dots = 0,$$

or we have

$$\frac{dR}{da} : \frac{dR}{db} : \&c. = A : B : \&c.,$$

a system of equations satisfied by the values x, y which belong to the common root.

In the case of a single equation $U = 0$ having a double root, the condition for this is $\Delta = 0$, where Δ is the discriminant of the function U ; and the like reasoning shows that for the values x, y which belong to the double root we have

$$\frac{dU}{da} : \frac{dU}{db} : \&c. = \frac{d\Delta}{da} : \frac{d\Delta}{db} : \dots;$$

viz. if U is of the form $(a, b, \dots)(x, y)^m$ with arbitrary coefficients, then we have thus a series of equations giving the required value of $\frac{x}{y}$; but if $(a, b, \dots)$ are arbitrary coefficients contained in any manner whatever in the function U , then we have a series of equations satisfied by the values x, y which belong to the double root.

9. *The normal at each point of a principal section of an ellipsoid is intersected by the normal at a consecutive point not on the principal section: show that the locus of the point of intersection is an ellipse having four (real or imaginary) contacts with the evolute of the principal section.*

The principal section is for convenience taken to be that in the plane of zx ; the coordinates of any point thereof are therefore $X, 0, Z$ where

$$\frac{X^2}{a^2} + \frac{Z^2}{c^2} = 1.$$

Consider the normal at a point X, Y, Z of the ellipsoid; taking x, y, z as current coordinates, the equations of the normal are

$$\frac{x - X}{X/a^2} = \frac{y - Y}{Y/b^2} = \frac{z - Z}{Z/c^2}.$$

Writing herein $y = 0$, we have

$$x = X \left(1 - \frac{b^2}{a^2}\right), \quad z = Z \left(1 - \frac{b^2}{c^2}\right);$$

viz. x, z are here the coordinates of the point where the normal meets the plane of xz ; and observing that the point in question lies on the normal at the point $X, 0, Z$, it is clear that x, y, z will be the coordinates of the intersection of the last-mentioned normal by the normal at the consecutive point not on the principal section.

Writing for shortness

$$\alpha = b^2 - c^2, \quad \beta = c^2 - a^2, \quad \gamma = a^2 - b^2,$$

($\alpha + \beta + \gamma = 0$, α and γ positive, β negative) the values are

$$x = \frac{\gamma X}{a^2}, \quad z = -\frac{\alpha Z}{c^2},$$

wherefore

$$\frac{X}{a} = \frac{ax}{\gamma}, \quad \frac{Z}{c} = -\frac{cz}{\alpha};$$

or, substituting in

$$\frac{X^2}{a^2} + \frac{Z^2}{c^2} = 1,$$

we have

$$\frac{a^2x^2}{\gamma^2} + \frac{c^2z^2}{\alpha^2} = 1,$$

the required locus, which is thus an ellipse.

If the point $(X, 0, Z)$ is an umbilicus, it is clear that the corresponding point of the locus will be a point of the evolute of the principal section; and to prove that the locus touches the evolute, it is only necessary to show that the tangent of the locus is also the tangent of the evolute; or what is the same thing, that the tangent of the locus passes through the umbilicus.

Now for the umbilicus we have

$$X^2 = -a^2\frac{\gamma}{\beta}, \quad Z^2 = -c^2\frac{\alpha}{\beta};$$

the corresponding values of x, z being

$$x = \frac{\gamma X}{a^2}, \quad z = -\frac{\alpha Z}{c^2}.$$

Take ξ, ζ as the current coordinates of a point on the tangent of the locus, we have

$$\frac{a^2x\xi}{\gamma^2} + \frac{c^2z\zeta}{\alpha^2} = 1,$$

or, substituting for x, z the foregoing values,

$$\frac{X\xi}{\gamma} - \frac{Z\zeta}{\alpha} = 1,$$

and these should be satisfied by $\xi, \zeta = X, Z$; viz. we ought to have

$$\frac{X^2}{\gamma} - \frac{Z^2}{\alpha} = 1,$$

and this equation is in fact true for the values of X, Z at the umbilicus; viz. for these values we have

$$-\frac{a^2}{\beta} + \frac{c^2}{\beta} = 1,$$

that is, $\beta = c^2 - a^2$, which is in fact the value of β .

There is obviously a contact in each quadrant, that is, there are four contacts (in the present case all real) of the locus with the evolute.

The same theorem holds good in regard to the other principal sections; only for these, the umbilici being imaginary, the points of contact of the locus with the evolute are also imaginary.

Remark. There is a great convenience in questions relating to the ellipsoid, in the use of the foregoing notations α, β, γ .

10. *An endless heavy chain of given length is suspended from two fixed points in the same horizontal plane: show that (subject to a condition as to the length) the figure of equilibrium may consist of portions of two distinct catenaries.*

The two parts of the chain will each of them be a portion of a catenary, viz. they will either coincide with each other, forming a twice repeated portion of a catenary (which is always a possible position of equilibrium), or they will form portions of two distinct catenaries. That the latter form is in some cases possible, appears from the case of a very long chain. It is then clear that there is a position of equilibrium in which the upper catenary is nearly a straight line. It may be added, that, as the length of the chain diminishes, the two distinct catenaries approach more and more, and for a certain value of the length become coincident; for any smaller value of the length, the only position is that consisting of a twice repeated portion of a catenary. But to obtain the solution in a regular manner, observe that, in order to the existence of such a form of equilibrium, the necessary condition is, that the tension at A (or B) must be equal in the two catenaries. Now the tension at any point of a catenary is proportional to the height above the directrix of the catenary; hence the condition is, that there shall be through the points A, B two catenaries having the same directrix, and such that the sum of the lengths is equal to the given length of the chain.

Take $AB = 2a$, the length of the chain $= 2l$. Take β for the distance of the directrix below the points A, B ; c for the parameter of the catenary (or distance of its lowest point above the directrix), β, c being of course unknown. Then taking the origin at the mid-point of the directrix, and the axis of y vertically upwards, the equation of the catenary is

$$y = \frac{c}{2} \left(e^{\frac{x}{c}} + e^{-\frac{x}{c}} \right),$$

whence for the point A or B ,

$$\beta = \frac{c}{2} \left(e^{\frac{a}{c}} + e^{-\frac{a}{c}} \right),$$

and the arc measured from the lowest point is

$$s = \frac{c}{2} \left(e^{\frac{x}{c}} - e^{-\frac{x}{c}} \right).$$

Hence, assuming that there are two distinct catenaries, if the parameters are c, c' , we have

$$\frac{1}{2}c \left(e^{\frac{a}{c}} + e^{-\frac{a}{c}} \right) = \frac{1}{2}c' \left(e^{\frac{a}{c'}} + e^{-\frac{a}{c'}} \right),$$

$$\frac{1}{2}c \left(e^{\frac{a}{c}} - e^{-\frac{a}{c}} \right) + \frac{1}{2}c' \left(e^{\frac{a}{c'}} - e^{-\frac{a}{c'}} \right) = l,$$

which are the conditions for the determination of c, c' ; and it is to be shown that these can be satisfied otherwise than by taking $c = c'$.

Trace the two curves

$$y = \frac{x}{2} \left(e^{\frac{a}{x}} + e^{-\frac{a}{x}} \right), \quad y' = \frac{x}{2} \left(e^{\frac{a}{x}} - e^{-\frac{a}{x}} \right),$$

shown respectively by the black line and the dotted line in fig. 2. Draw any line parallel to the axis of x , meeting the first curve in the points P, P' respectively, and let the ordinates $MP, M'P'$ meet the second curve in the points Q, Q' respectively;

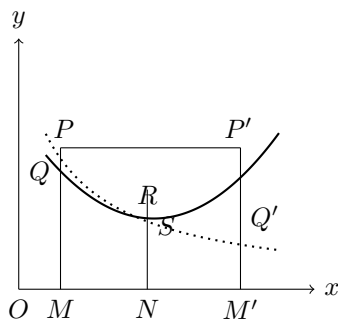


Fig. 2.

then it is clear, that if for a given value of l the line PP' can be drawn in such-wise that $MQ + M'Q' = l$, there will be in fact the required two values $c = OM$ and $c' = OM'$.

And since for MP very large we have MQ , and therefore also $MQ + M'Q'$, very large, and as MP diminishes, $MQ + M'Q'$ also diminishes until it attains a certain minimum value, say $= \lambda$, it is clear that if l has any value greater than this minimum value, PP' can be so drawn that $QM + Q'M' = l$.

[The above remarkably elegant investigation in regard to the two values c, c' was given in the Examination; it seems to be the case that as PP' moves downwards, $MQ + M'Q'$ continually decreases (viz. MQ decreases more rapidly than $M'Q'$ increases), its value being least, and $= 2NS$ when PP' becomes a tangent to the first curve at its lowest point R ; but it is not by any means easy to prove that this is so. The question depends on the form of the curve defined by the equations

$$X = \frac{1}{2}x_1 \left(e^{\frac{a}{x_1}} - e^{-\frac{a}{x_1}} \right) + \frac{1}{2}x_2 \left(e^{\frac{a}{x_2}} - e^{-\frac{a}{x_2}} \right),$$

$$Y = \frac{1}{2}x_1 \left(e^{\frac{a}{x_1}} + e^{-\frac{a}{x_1}} \right) = \frac{1}{2}x_2 \left(e^{\frac{a}{x_2}} + e^{-\frac{a}{x_2}} \right),$$

where X and Y are the current coordinates.]

11. A particle is attracted to two centres of force, one of them at the origin, the other revolving about the origin in a circle in the plane of xy with a uniform angular velocity n' : find the equations of motion; and writing v for the velocity of the particle and A for the resolved area (about the fixed centre) in the plane of xy , show that there is a first integral giving the value of $v^2 - 4n' \frac{dA}{dt}$ in terms of the coordinates of the particle and of the revolving centre.

Take $\xi, \eta, 0$ for the coordinates of the moving centre, so that

$$\xi = a \cos n't, \quad \eta = a \sin n't;$$

the equations of motion are

$$\begin{aligned} \frac{d^2x}{dt^2} &= -\phi r \frac{x}{r} - \psi \rho \frac{x - \xi}{\rho}, \\ \frac{d^2y}{dt^2} &= -\phi r \frac{y}{r} - \psi \rho \frac{y - \eta}{\rho}, \\ \frac{d^2z}{dt^2} &= -\phi r \frac{z}{r} - \psi \rho \frac{z}{\rho}, \end{aligned}$$

where

$$r^2 = x^2 + y^2 + z^2, \quad \rho^2 = (x - \xi)^2 + (y - \eta)^2 + z^2.$$

We have

$$\begin{aligned} r \, dr &= x \, dx + y \, dy + z \, dz, \\ \rho \, d\rho &= (x - \xi)(dx - d\xi) + (y - \eta)(dy - d\eta) + z \, dz. \end{aligned}$$

But

$$d\xi = -n'a \sin n't \, dt = -n'\eta \, dt, \quad d\eta = n'a \cos n't \, dt = n'\xi \, dt,$$

whence

$$\begin{aligned} \rho \, d\rho &= (x - \xi)dx + (y - \eta)dy + z \, dz \\ &\quad + n'\{\eta(x - \xi) - \xi(y - \eta)\}dt \\ &= (x - \xi)dx + (y - \eta)dy + z \, dz \\ &\quad - n'\{x(y - \eta) - y(x - \xi)\}dt. \end{aligned}$$

Hence from the equations of motion

$$\begin{aligned} &2 \left(\frac{dx}{dt} \frac{d^2x}{dt^2} + \frac{dy}{dt} \frac{d^2y}{dt^2} + \frac{dz}{dt} \frac{d^2z}{dt^2} \right) - 2n' \left(x \frac{d^2y}{dt^2} - y \frac{d^2x}{dt^2} \right) \\ &= -\frac{\phi r}{r} 2 \left(x \frac{dx}{dt} + y \frac{dy}{dt} + z \frac{dz}{dt} \right) \\ &\quad - \frac{\psi \rho}{\rho} \left[2 \left\{ (x - \xi) \frac{dx}{dt} + (y - \eta) \frac{dy}{dt} + z \frac{dz}{dt} \right\} \right. \\ &\quad \left. - 2n' \{x(y - \eta) - y(x - \xi)\} \right]. \end{aligned}$$

But we have

$$v^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2,$$

$$x \frac{d^2 y}{dt^2} - y \frac{d^2 x}{dt^2} = \frac{d}{dt} r^2 \frac{d\theta}{dt} = 2 \frac{dA}{dt};$$

the foregoing equation may be written

$$\frac{d.v^2}{dt} - 4n' \frac{d^2 A}{dt^2} = -2\phi r \frac{dr}{dt} - 2\psi \rho \frac{d\rho}{dt},$$

whence

$$v^2 - 4n' \frac{dA}{dt} = C - 2 \int \phi r dr - 2 \int \psi \rho d\rho,$$

the required result.

12. If x, y are the coordinates of a particle moving in plano under the action of a central force varying as (distance)⁻²: write down the expressions of the coordinates x, y in terms of the time t and of four arbitrary constants: and (in case of disturbed motion) starting from the equations

$$\delta x = 0, \quad \delta y = 0, \quad \delta x' = \frac{d\Omega}{dx} dt, \quad \delta y' = \frac{d\Omega}{dy} dt,$$

(the notation to be explained), indicate the process of finding the variations of the constants in terms of (1) $\frac{d\Omega}{dx}, \frac{d\Omega}{dy}$, (2) the derived functions of Ω in regard to the constants.

We have

$$x = a \left\{ \frac{\cos u - e}{1 - e \cos u} \cos \varpi + \frac{\sqrt{1 - e^2} \sin u}{1 - e \cos u} \sin \varpi \right\},$$

$$y = a \left\{ \frac{\sqrt{1 - e^2} \sin u}{1 - e \cos u} \cos \varpi - \frac{\cos u - e}{1 - e \cos u} \sin \varpi \right\},$$

where

$$u - e \sin u = t \sqrt{\frac{\mu}{a^3}} + c,$$

an equation serving to express u in terms of t and the constants a, e, c ; the foregoing equations, therefore, in effect give x, y in terms of t and the four constants a, e, c, ϖ .

In the second part of the question, Ω is a given function of x, y, t , the differential coefficients $\frac{d\Omega}{dx}, \frac{d\Omega}{dy}$ being the partial ones in regard to x, y respectively. The equation $\delta x = 0$ signifies that the variation of x , in so far as it arises from the variation of the constants, is $= 0$; it in fact means

$$\frac{dx}{da} \frac{da}{dt} + \frac{dx}{de} \frac{de}{dt} + \frac{dx}{dc} \frac{dc}{dt} + \frac{dx}{d\varpi} \frac{d\varpi}{dt} = 0.$$

The value of $x' (= \frac{dx}{dt})$ is therefore obtained from that of x by differentiating in regard to t alone, as if a, e, c, ϖ were constants: viz. x' will be a given function of t, a, e, c, ϖ ; $\delta x'$ then denotes the variation of x' in so far as it arises from the variation of the constants, viz. the equation $\delta x' = \frac{d\Omega}{da} dt$ means

$$\frac{dx'}{da} \frac{da}{dt} + \frac{dx'}{de} \frac{de}{dt} + \frac{dx'}{dc} \frac{dc}{dt} + \frac{dx'}{d\varpi} \frac{d\varpi}{dt} = \frac{d\Omega}{dx}.$$

There are the like equations in regard to y, y' , viz. in all, four equations which are linear in regard to $\frac{da}{dt}, \frac{de}{dt}, \frac{dc}{dt}, \frac{d\varpi}{dt}$; and which serve to determine these quantities in terms of $\frac{d\Omega}{dx}, \frac{d\Omega}{dy}$.

Now considering the x, y as expressed in terms of a, e, c, ϖ, t , then Ω becomes a function of these quantities; the differential coefficients $\frac{d\Omega}{da}$, &c., being connected with the original differential coefficients $\frac{d\Omega}{dx}, \frac{d\Omega}{dy}$ by the equations

$$\frac{d\Omega}{da} = \frac{d\Omega}{dx} \frac{dx}{da} + \frac{d\Omega}{dy} \frac{dy}{da},$$

$$\frac{d\Omega}{de} = \frac{d\Omega}{dx} \frac{dx}{de} + \frac{d\Omega}{dy} \frac{dy}{de},$$

&c.

As there are four equations, $\frac{d\Omega}{dx}, \frac{d\Omega}{dy}$ can be expressed in an infinity of ways in terms of $\frac{d\Omega}{da}, \frac{d\Omega}{de}, \frac{d\Omega}{dc}, \frac{d\Omega}{d\varpi}$; and considering $\frac{da}{dt}$, &c., as given in terms of $\frac{d\Omega}{dx}, \frac{d\Omega}{dy}$, we can in an infinity of ways express $\frac{da}{dt}$, &c., as linear functions of $\frac{d\Omega}{da}, \frac{d\Omega}{de}, \frac{d\Omega}{dc}, \frac{d\Omega}{d\varpi}$. But there is one form (obtained by combining the equations in a particular manner) wherein the coefficients of the last-mentioned quantities are functions of a, e, c, ϖ without t ; and this is the form actually employed for the expression of $\frac{da}{dt}, \frac{de}{dt}, \frac{dc}{dt}, \frac{d\varpi}{dt}$ in terms of $\frac{d\Omega}{da}, \frac{d\Omega}{de}, \frac{d\Omega}{dc}, \frac{d\Omega}{d\varpi}$, in the method wherein these quantities are made use of.

I remark upon the present question, that the answer ought to be in substance perfectly familiar to every student in Physical Astronomy; and that a student ought to be able to present it in a clear and logical form: the question being in fact intended as a test of ability in this respect.

13. *Explain the course of the geodesic lines on a spheroid of revolution: and in particular show that the condition is satisfied in virtue of which any geodesic line, considered as starting from a given point, ceases at some point of its course to be a shortest line.*

From each point on the surface a geodesic line may be drawn in any direction whatever along the surface, that is, through each point of the surface there is a singly infinite series of geodesic lines. A geodesic line undulates (in the manner of a sinusoid) between two parallels equidistant from the equator on opposite sides thereof; viz. considering it as starting from a point A on the equator, it arrives at a point V on the upper parallel (there touching the parallel), and passes downwards to cut the equator at A' , and thence arrives at a point V' on the lower parallel (there touching the parallel), and again passes upwards to meet the equator at A'' , and so on; the arcs $AV, VA', A'V', V'A'',$ &c., being similar and equal to each other (differing only in position). The equatoreal arc $AA' (= A'A'' = \&c.)$ or difference of the longitudes A, A' , is always less than 180° , its value increasing with the inclination at which the geodesic line cuts the equator, viz. when this angle is indefinitely small, the arc is $= \frac{c}{a}180^\circ$ (c, a the polar and equatoreal axes respectively), and as the inclination becomes indefinitely near to 90° , the value of the arc becomes indefinitely near 180° . If the arc in question is commensurable with 180° , the geodesic line will be, it is clear, a closed curve; but if not, then it is not a closed curve, but proceeds undulating for ever between the two parallels. In the limiting case where the inclination is $= 90^\circ$, the geodesic line is obviously a meridian.

Considering a geodesic line starting in a given direction from a point A , and the geodesic line from the same point A in the consecutive direction, it appears from the foregoing account of the configuration of the lines, that the two lines will intersect each other in general an indefinite number of times: supposing that they first intersect in a point K , then by a general theorem of Jacobi's, the geodesic line AK is a shortest line from A to any point nearer than K , but it is not a shortest line from A to any point beyond K .